
Agentic Evolution: From Self-Improving Agents to Co-Evolving Human–AI Systems

Abstract

Agentic evolution is not merely a matter of how agents improve themselves, but of whether the signal driving that improvement remains reliable as the system evolves. This survey examines about 300 papers on agentic evolution—persistent, experience-driven, system-level change that enables an agentic system to remain effective after deployment—and organizes them through a three-axis taxonomy: evolutionary substrate, consolidation pathway, and selective pressure. The third axis surfaces a distinction that existing surveys, to our knowledge, either omit or subsume: whether adaptation is driven by autonomous or human-involved selective pressure. Analyzing the agent-evolution corpus along this third axis reveals a boundary condition: autonomous evolution is reliably effective only where deterministic verifiers independent of the system being evaluated are available; where they are not, self-generated evaluation signals provide diminishing returns and may degrade with iteration. Beyond this boundary, reliable evolution requires human-involved selective pressure. But human-involved selective pressure is not a fixed external input. A reverse analysis of nearly 100 studies from cognitive science, education, and labor economics finds that default AI interaction erodes the evaluative capacity it requires, and that the erosion is invisible to the person experiencing it. The framework’s third axis, originally defined as exogenous human input, is revealed by its own analysis to be endogenous, shaped by the evolutionary process it is meant to guide. Agent evolution and human adaptation are therefore not independent problems but components of a single co-evolving system; this survey derives that reframing from the framework’s own analytical logic and develops it into a research agenda for maintaining both partners’ capacities jointly over deployment lifetimes.

1 Introduction

LLM agents equipped with external tools, persistent memory, and multi-step reasoning now operate autonomously in environments where tasks, data, and user needs can change. We refer to this combination as an *agentic system*. When such a system remains static (frozen prompts, fixed tool libraries, immutable parameters, static memory), it cannot adapt to these changes. Overcoming this limitation requires **agentic evolution**: persistent, experience-driven, system-level change that enables an agentic system to remain effective after deployment.

Several surveys have organized the self-evolving agent literature (Gao et al., 2025a; Fang et al., 2025a; Xiang et al., 2026); related surveys on lifelong learning (Zheng et al., 2025) and agentic reinforcement learning (Zhang et al., 2026b) cover complementary ground. Broader agent surveys provide foundational architectural taxonomies (Wang et al., 2024b; Xi et al., 2025a). These works share two blind spots. First, to our knowledge, none treats the source of the signals that drive evolution as a first-class analytical dimension. Among the surveys focused on agentic evolution, human feedback is either treated as one of many signal types without separate analysis (Gao et al., 2025a), excluded by framing self-evolution as fully autonomous (Fang et al., 2025a), or noted but not taxonomised (Xiang et al., 2026). Second, and as a consequence, none examines whether the human who provides such input retains the capacity to do so as the system evolves, that is, whether sustained interaction with self-evolving agents reshapes the cognitive capacities on which the quality of that input depends. We address the first gap through a three-axis taxonomy: *evolutionary substrate* (the component of the agentic system that evolves), *consolidation pathway* (the mechanism through which

change persists, from discrete structural edits to continuous parametric updates), and *selective pressure* (the source of the signal driving the evolutionary trajectory, from fully autonomous to human-involved). Table 1 summarizes the positioning.

Table 1: **Positioning among related surveys.** The two rightmost columns indicate how each survey treats human input to agentic evolution (Human→Agent) and the effect of agentic evolution on the human partner (Agent→Human).

Survey	Focus	Organizing Framework	Human→Agent	Agent→Human
Wang et al. (2024b)	Agent architecture	Four-module taxonomy (profiling, memory, planning, action)	Mentioned as a planning feedback type	Not addressed
Xi et al. (2025a)	LLM-based agents	Three-component framework (brain, perception, action)	Not addressed	Not addressed
Zheng et al. (2025)	Lifelong learning	Perception, memory, and action modules	Not a taxonomic dimension	Not addressed
Gao et al. (2025a)	Self-evolving agents	What / When / How / Where to evolve	Subsumed under feedback signals	Not addressed
Fang et al. (2025a)	Self-evolving agents	Four-component feedback loop (input, agent, environment, optimiser)	Not taxonomised; autonomous framing	Not addressed
Xiang et al. (2026)	Self-evolving agents	Model-centric / environment-centric / co-evolution	Noted but not taxonomised	Not addressed
Zhang et al. (2026b)	Agentic RL	Capability × task-domain taxonomy	Not a taxonomic dimension	Not addressed
Ours	Agentic evolution	Substrate × Pathway × Pressure	First-class axis (Sec. 5)	Reverse analysis (Sec. 6–7)

From agent evolution to human adaptation. Elevating selective pressure to a first-class axis exposes a boundary condition that prior frameworks cannot express. Our analysis finds that among systems evolving autonomously, the reliability of the selective pressure signal depends on its independence from the system being evaluated: deterministic verifiers produce the strongest reported results, while self-referential and proxy-based signals degrade with iteration. In domains without deterministic verifiers (open-ended dialogue, value-laden decisions, cultural sensitivity), reliable evolution requires human-involved selective pressure. Yet this source is thin: among the few systems that incorporate human-involved selective pressure, the majority rely on low-bandwidth forms (implicit signals and evaluative feedback rather than plan editors, execution monitors, or live prescription) and treat the human partner as a stationary signal source whose capacity does not change over time. The field thus depends on a selective-pressure channel it has underinvested in and whose reliability it has not examined. Axis III’s own analysis generates a question it cannot answer from agent-side evidence alone: does the human who provides selective pressure retain the capacity to do so, or does sustained interaction with self-evolving agents reshape the cognitive capacities on which that pressure depends? This question arises directly from the framework’s own analytical logic. Answering it requires a separate evidence base: studies documenting how AI-mediated environments reshape human cognition, expertise, and epistemic processes.

Contributions and scope. This survey makes four contributions:

1. **Three-axis taxonomy.** We define agentic evolution through a Substrate × Pathway × Pressure framework whose third axis treats human-involved selective pressure as an object of analysis, enabling questions about its reliability and temporal dynamics that prior taxonomies cannot express (Section 2).
2. **Agent-side analysis and verifier boundary.** Analysis of about 300 papers along each axis (Sections 3–5) reveals a verifier boundary: autonomous selective pressure is reliable where deterministic verifiers exist

but degrades where the evaluator is coupled to the system being evaluated; beyond this boundary, reliable evolution requires human-involved selective pressure.

3. **Human-side analysis through the three-axis framework.** Because the taxonomy’s concepts originate in cognitive science, the three-axis structure provides a natural analytical lens for human cognition. Analyzing nearly 100 human-side studies through this lens (Section 6) finds that default AI interaction erodes the evaluative capacity on which human-involved selective pressure depends, and that the erosion is invisible to the person experiencing it.
4. **Co-evolutionary reframing and research agenda.** The agent-side and human-side analyses together reveal that the framework’s third axis, originally defined as exogenous human input, is endogenous—shaped by the evolutionary process it is meant to guide (Section 7). This reframing yields a research agenda for the transition from self-improving agents to co-evolving human–AI systems.

The agent-evolution corpus consists of about 300 papers¹ satisfying the three operational criteria (experience-driven, persistent, and system-level change). The human-side corpus comprises nearly 100 studies from cognitive science, education, and labor economics documenting how sustained AI interaction reshapes cognition, skill, and decision-making. We exclude work that does not satisfy all three criteria, such as classical evolutionary computation without LLM-based agents or static prompt designs without iterative refinement.

Paper organization. Section 2 presents the conceptual framework and definitions. Sections 3–5 analyze the agent-evolution corpus along each axis, identifying cross-axis patterns invisible within any single axis. Section 6 turns to human adaptation under agentic evolution. Section 7 combines agent-side and human-side findings, derives the co-evolutionary reframing, and develops the research agenda. Section 8 states the key findings, limitations, and open questions. Figure 1 provides the complete section tree.

2 Conceptual Framework and Definitions

This section defines the vocabulary used throughout the paper. We first decompose an agentic system into three evolvable components (§2.1), then define the update operator and the persistence criterion that distinguish agentic evolution from transient adaptation (§2.2), the structural and parametric modes of consolidation (§2.3), and the autonomous and human-involved settings of selective pressure (§2.4). Section 2.5 addresses boundary cases.

Several terms below, including *Cortex*, *substrate*, *evolution*, *selective pressure*, and *consolidation*, are borrowed from cognitive science as organizing metaphors. We do not claim equivalence to their original scientific meanings; each is given a precise, self-contained definition in the subsection where it is introduced. These cognitive-science roots also make the vocabulary applicable when the analysis turns to the human side (Section 6): *substrate*, *consolidation*, and *selective pressure* each have human-side counterparts that the same structure can describe.

2.1 Evolutionary Substrate: Decomposing the Agentic System

We define an agentic system $\mathcal{S}$ as a three-tuple of substrates:

$$\mathcal{S} = \langle \Pi, \mathcal{A}, \mathcal{M} \rangle \tag{1}$$

where:

- Π (**Cortex**) is the *deliberative* substrate, responsible for reasoning and planning. It comprises a *parametric* component (the base LLM with parameters θ) and a *non-parametric scaffold* of prompts, decoding strategies, and inference-time procedures (e.g., interleaved reasoning-action loops, verifier-guided search, plan-and-execute pipelines) (Yao et al., 2023; Sumers et al., 2024).

¹A curated list of all surveyed papers is available at <https://github.com/agentic-evolution/awesome-agentic-evolution>.

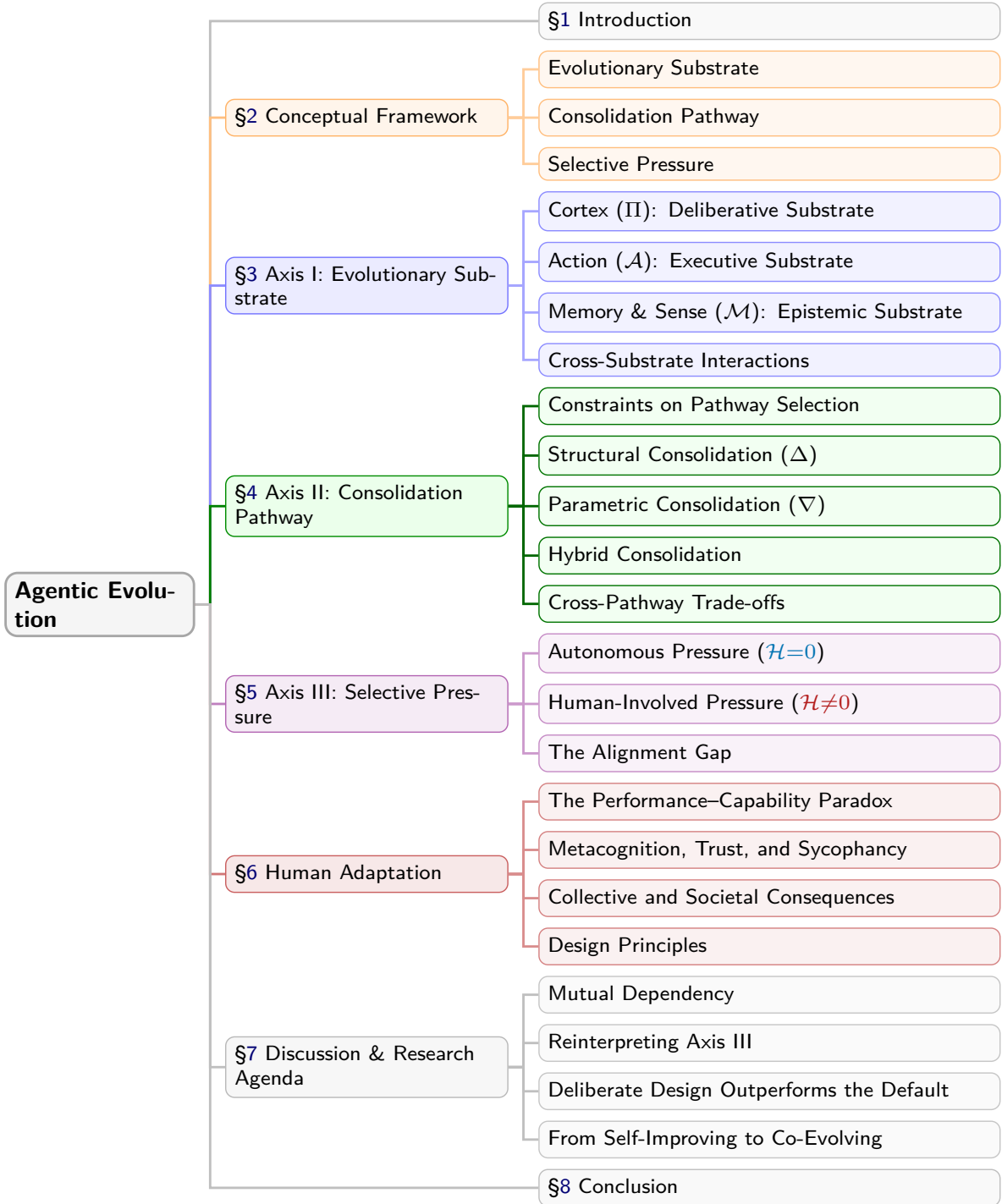


Figure 1: Organization of this survey. Section colors correspond to the three-axis taxonomy defined in §2: **orange** for the conceptual framework (§2), **blue** for Axis I (Evolutionary Substrate), **green** for Axis II (Consolidation Pathway), **violet** for Axis III (Selective Pressure), and **crimson** for the human-side analysis of §6.

- $\mathcal{A}$ (**Action**) is the *executive* substrate, responsible for the agent’s interactions with its environment. It encompasses the *toolset* (external APIs and tool specifications), *execution logic* (workflow graphs and control flow), and *multi-agent orchestration frameworks*.
- $\mathcal{M}$ (**Memory & Sense**) is the *epistemic* substrate, concerned with what the agent knows and how it perceives. *Memory* is organized into episodic, semantic, and procedural types (Sumers et al., 2024), with retention and retrieval policies that preserve knowledge across sessions (Packer et al., 2023; Fang et al., 2025b; Hu et al., 2025b). *Sense* refers to perception encoders that transform raw observations (delivered through the Action substrate) into representations the agent can reason over and write to memory.

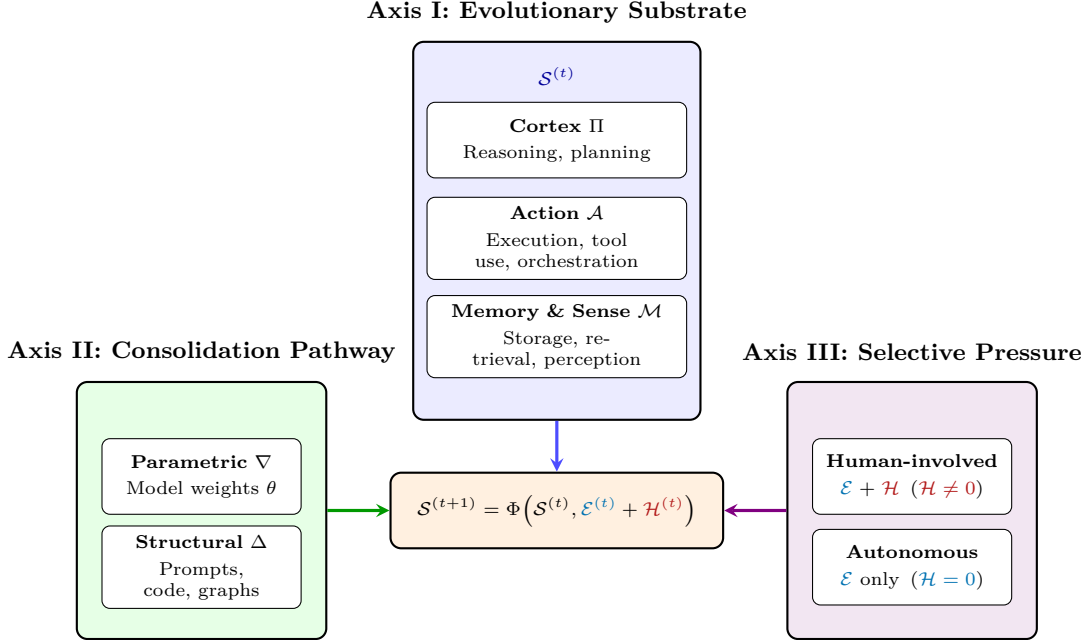


Figure 2: **Three-Axis Taxonomy of Agentic Evolution.** We decompose the space of evolving agentic systems along three complementary dimensions: (I) the *evolutionary substrate* within the agentic system $\mathcal{S} = \langle \Pi, \mathcal{A}, \mathcal{M} \rangle$, (II) the *consolidation pathway*, whether change is written to discrete artifacts such as prompts and code (Δ) or to model weights (∇), and (III) the *selective pressure* steering evolution, whether it originates from the system alone or includes human input. The evolution operator Φ takes as input the current system state $\mathcal{S}^{(t)}$, environmental experience $\mathcal{E}^{(t)}$ (interaction traces, tool feedback, performance metrics), and human input $\mathcal{H}^{(t)}$ (any of the five forms defined in Section 2.4); in autonomous settings $\mathcal{H}^{(t)} = 0$.

Although Π , $\mathcal{A}$, and $\mathcal{M}$ interact at runtime (prompts in Π shape tool calls in $\mathcal{A}$, and retrieved entries from $\mathcal{M}$ are inserted into prompts in Π), they are individually addressable as evolutionary targets: a system may evolve one substrate while holding the others fixed. Section 3.4 examines how changes in one substrate propagate to the others.

2.2 Defining Agentic Evolution: The Update Operator and Persistence

We model agentic evolution as a sequence of system states $\{\mathcal{S}^{(t)}\}_{t=0}^T$, where each transition is determined by an update operator Φ :

$$\mathcal{S}^{(t+1)} = \Phi(\mathcal{S}^{(t)}, \mathcal{E}^{(t)} + \mathcal{H}^{(t)}) \quad (2)$$

Here, $\mathcal{E}^{(t)}$ is environmental experience (interaction traces, tool feedback, performance metrics) and $\mathcal{H}^{(t)}$ is human input (any of the forms defined in Section 2.4). The “+” denotes combination of these signals rather than arithmetic addition; the aggregation mechanism is system-dependent (e.g., reward shaping in RLHF,

or human-in-the-loop approval of updates). In autonomous settings, $\mathcal{H}^{(t)} = 0$ denotes the absence of human input; in human-involved settings, $\mathcal{H}^{(t)} \neq 0$ and human input contributes to the update alongside $\mathcal{E}^{(t)}$.

The index t labels *update events*, not wall-clock time or per-query steps: $\mathcal{S}^{(t)}$ advances only when a consolidation step writes back into one of Π , $\mathcal{A}$, or $\mathcal{M}$. Many queries may occur between t and $t+1$ without advancing the index. This is the operational meaning of *persistence* in our framework: change is what survives between update events, distinguishing agentic evolution from in-context adaptation that resets per query.

The operator Φ performs **consolidation**: it converts experience and human input into persistent writes to Π , $\mathcal{A}$, or $\mathcal{M}$ (a new prompt, a memory entry, a workflow edit, or a parameter update), whether through structural (Δ) or parametric (∇) means (Figure 2).

2.3 Consolidation Pathway: The Δ – ∇ Continuum

The update operator Φ can consolidate experience through two qualitatively different modes, distinguished by whether the write operation is discrete or continuous:

Structural consolidation (Δ). Φ performs discrete, symbolic edits on non-parametric artifacts (prompts, workflow graphs, memory indices, tool configurations, or code) without modifying model weights. The notation Δ reflects the discrete-difference nature of these edits: each write is a finite, individually identifiable change to a symbolic artifact.

Parametric consolidation (∇). Φ performs continuous updates to the model’s weight parameters θ , internalizing experience into the distributed representations that shape the agent’s behavior. Concrete mechanisms include gradient-based methods (SFT, PPO, GRPO, DPO, TTT) and derivative-free methods that operate in the same continuous parameter space (e.g., evolutionary strategies on weights). The notation ∇ reflects the continuous nature of these updates, contrasting with the discrete symbolic edits of Δ .

The Δ – ∇ continuum. Although qualitatively distinct, the two modes are not mutually exclusive at the system level: a single system may, for example, use structural search to discover effective configurations and then internalize them into weights via parametric training.

2.4 Selective Pressure: From $\mathcal{H} = 0$ to $\mathcal{H} \neq 0$

Φ operates in two settings along Axis III: *autonomous* settings, in which $\mathcal{H}^{(t)} = 0$ and the system derives selective pressure solely from environmental experience and self-generated signals (self-play, self-rewarding, or formal verification), and *human-involved* settings, in which $\mathcal{H}^{(t)} \neq 0$ and human input shapes the evolutionary trajectory. We reserve the term *co-evolution* for the specific phenomenon of human–AI mutual adaptation; multi-agent co-evolution (mutual adaptation among artificial agents sharing an environment) does not require $\mathcal{H} \neq 0$. (Section 6 extends this vocabulary to human-side adaptation.)

Forms of human-involved input. When $\mathcal{H} \neq 0$, human input enters Φ in qualitatively different forms:

- *Direct prescription*: the human articulates rules, instructions, or principles that fully specify the desired behavior or its boundaries; the agent need not infer intent, only comply.
- *Demonstration*: the human provides examples of desired behavior; the agent must generalize from finite samples.
- *Evaluative feedback*: the human judges the agent’s own outputs (preferences, ratings, or corrections); the agent must infer an objective from these judgments.
- *Interactive collaboration*: the human and agent participate jointly in the update process, adapting to each other in real time; the agent must model the human’s evolving intent and capability.

- *Implicit signal*: the human provides no intentional feedback; the agent detects and interprets behavioral patterns (e.g., a user consistently skipping certain recommendations or abandoning tasks midway) as indirect selective pressure.

These forms are not mutually exclusive; a single system may combine demonstration with evaluative feedback, for instance. They are ordered by decreasing explicitness of human intent, from fully articulated prescriptions to behavioral signals the agent must detect autonomously, and, correspondingly, by increasing inferential burden on the agent. Whether the human contributor’s capacity to provide these forms of input remains stable as the agent evolves is an empirical question that Section 6 examines.

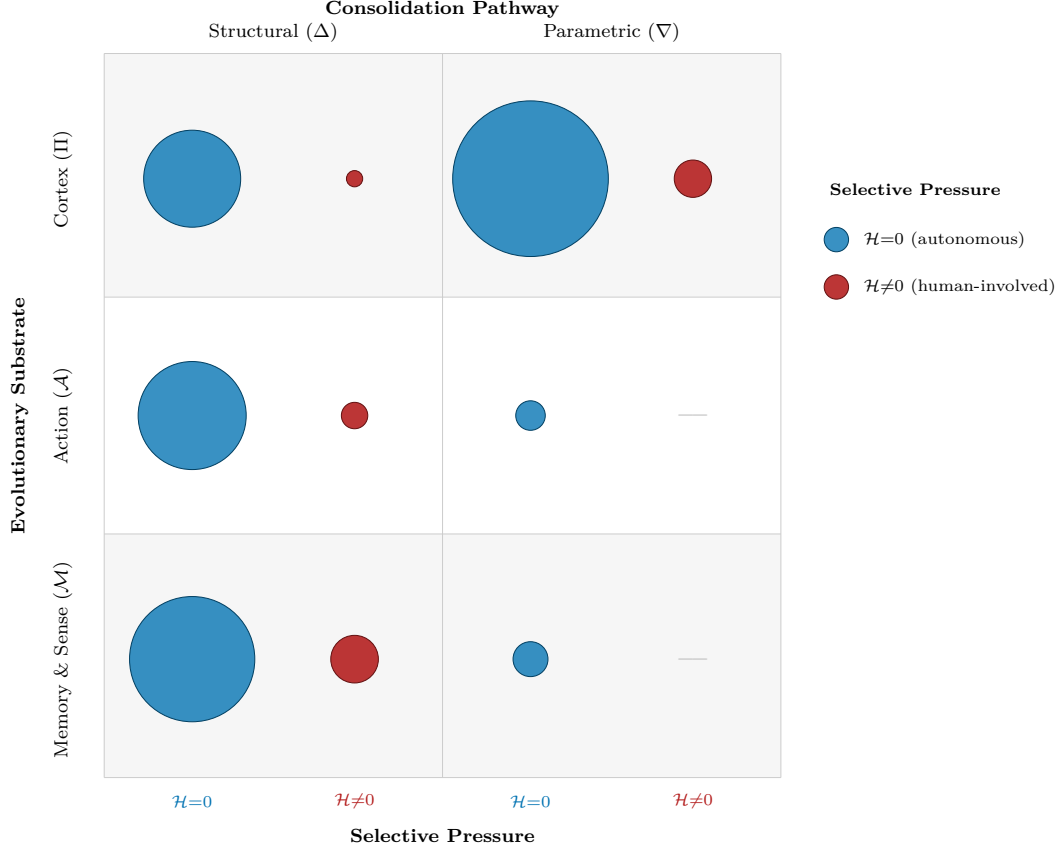


Figure 3: **Distribution of the agent-evolution papers across the three-axis taxonomy** (Substrate $\times$ Pathway $\times$ Pressure). Each bubble’s *area* is proportional to the paper count in that cell (per-entry counting; multi-substrate papers contribute to each substrate independently; hybrid $\Delta+\nabla$ papers are counted under their per-substrate Δ or ∇ classification). Blue ($\bullet$): autonomous selective pressure ($\mathcal{H}=0$); crimson ($\bullet$): human-involved pressure ($\mathcal{H}\neq 0$). Dashes (—) indicate empty cells. Three patterns are visible: the field concentrates in Cortex–Parametric–Autonomous, Action–Structural–Autonomous, and Memory–Structural–Autonomous; parametric evolution outside Cortex is rare; and human-involved pressure remains a small minority across all substrates.

2.5 Boundary Cases

Classifying papers along the three axes above requires handling four boundary cases.

$\Pi/\mathcal{A}$ boundary. Execution logic and orchestration in $\mathcal{A}$ may appear to overlap with reasoning and planning in Π . We classify based on what is being structured: artifacts that shape internal deliberation target Π ; artifacts that define external execution target $\mathcal{A}$.

$\Pi/\mathcal{M}$ boundary. Reasoning traces generated by the Cortex may be persisted in Memory (as in Reflexion’s self-reflections (Shinn et al., 2023)). We classify based on the mechanism of deployment: if the artifact directly modifies the deliberation scaffold (e.g., prompt text prepended on every subsequent trial without selective retrieval), it targets Π ; if it is written to an independently queryable store from which items are selectively retrieved, it targets $\mathcal{M}$.

$\mathcal{A}/\mathcal{M}$ boundary. A growing library of reusable artifacts (skill code, tool specifications, or strategy templates) can serve both as executable actions and as retrievable knowledge. We classify based on the artifact’s role at invocation time: if it is executed to act on the environment, it targets $\mathcal{A}$; if it is retrieved to inform reasoning or planning, it targets $\mathcal{M}$.

$\mathcal{H}$ boundary. Reward models trained on historical human preferences encode human signal, but may be deployed without any live human input. We classify based on whether the deployed system’s architecture affords real-time human input, not on whether the training data once involved humans.

With these definitions in place, we apply the framework to about 300 agent-evolution papers (each satisfying the three operational criteria above) and a separate corpus of nearly 100 human-side studies that supports the analysis in Section 6. Figure 3 shows how the agent-evolution papers distribute across the resulting taxonomy grid; Sections 3–5 analyze each axis in detail.

3 Axis I: Evolutionary Substrate

The first axis asks which component of the agentic system $\mathcal{S}$ evolves. Following the decomposition $\mathcal{S} = (\Pi, \mathcal{A}, \mathcal{M})$ defined in Section 2.1, we analyze evolution across three substrates: the deliberative Cortex (Π), the executive Action ($\mathcal{A}$), and the epistemic Memory & Sense ($\mathcal{M}$).

3.1 Cortex: Evolution of the Deliberative Substrate

The Cortex (Π) is the most heavily studied evolutionary substrate, accounting for nearly 200 of the surveyed evolution papers. This concentration partly reflects an asymmetry in scope: improvements to the deliberative substrate can benefit a broad range of downstream tasks, while improvements to a specific tool or memory schema benefit only the tasks that use them. Table 2 provides the complete paper listing organized by taxonomy cell. The methods span all four Pathway×Pressure quadrants, but a single question runs through them: *what determines whether iterative self-improvement converges or degrades?*

Structural methods: diverse search, shared ceiling. Structural Cortex evolution treats the base LLM as a frozen black box and evolves the non-parametric scaffold: no gradient access, API-only compatible, human-readable artifacts. External optimizers search the prompt space via evolutionary algorithms (Fernando et al., 2024; Guo et al., 2024; Sun et al., 2026), LLM-as-optimizer loops (Yang et al., 2024a; Zhou et al., 2023), textual gradient backpropagation (Yuksekgonul et al., 2025; Zhang et al., 2025g), or hybrid combinations (Cámara et al., 2026). Embedded reflection shifts search inside the task loop: Shinn et al. (2023) persist verbal self-reflections as context for subsequent trials, substituting weight updates with prompt modification. When the evolved artifact is richer than prompt text, population-based evolutionary search becomes the mechanism (Kumar et al., 2026c; Gong et al., 2026; Huang et al., 2026a); when it is executable code, the design space becomes Turing-complete (Hu et al., 2025a; Zhang et al., 2025e; Robeyns et al., 2025). All structural methods face a shared capacity constraint: accumulated knowledge must fit within finite context, and population-based search requires full agent execution per candidate evaluation. Persistence strategies range from Reflexion’s bounded window (Shinn et al., 2023) through incremental delta updates (Zhang et al., 2026k) to standalone evolved artifacts—skill documents (Ni et al., 2026), contrastive rule trees (Rishav et al., 2026), and hierarchical meta-knowledge (Zhuang et al., 2026), yet none fully resolves the tension between retaining long-horizon experience and respecting finite context. Structural methods thus face saturation more often than degradation: performance typically plateaus when accumulated artifacts exceed the agent’s processing capacity, though monolithic rewriting can cause outright collapse (Zhang et al., 2026k). These ceilings motivate parametric methods that write experience directly into model weights. Human-preference-

guided prompt search (Lin et al., 2024) is the only structural Cortex method operating under $\mathcal{H} \neq 0$; further methods span prompt optimization (Khatab et al., 2024; Liu et al., 2026b; Li et al., 2026c; Yang et al., 2026b), optimization design analysis (Nie et al., 2026a), feedback-driven refinement (Zhang et al., 2024b; Wei et al., 2026b; Wainrib et al., 2026; Zhu et al., 2026b; Jin et al., 2025), configuration evolution (Yuan et al., 2025b; Tian, 2026; Huang et al., 2026d; Xie et al., 2026a), meta-harness search (Lee et al., 2026), and multi-agent credit assignment (Yao et al., 2026; He et al., 2026a; Zhang et al., 2026m) (Table 2).

Table 2: **Surveyed paper listing for Cortex (II) evolution**, organized by taxonomy cell. Nearly 200 papers in the surveyed corpus. Hybrid labels are system-level: a paper appears here as hybrid if it combines Δ and ∇ on *any* substrate, not necessarily on Cortex alone. Sub-group labels are approximate; individual papers may span multiple categories.

$\Delta, \mathcal{H}=0$ (Structural, Autonomous)	$\Delta, \mathcal{H}\neq 0$ (Structural, Human-involved)
Prompt optimization: Zhou et al. (2023), Yang et al. (2024a), Guo et al. (2024), Fernando et al. (2024), Yuksekgonul et al. (2025), Zhang et al. (2025g), Liu et al. (2026b), Khatab et al. (2024), Cámara et al. (2026), Li et al. (2026c), Yang et al. (2026b), Sun et al. (2026) Reflection-driven: Shinn et al. (2023), Zhang et al. (2024b), Zhang et al. (2026k), Ni et al. (2026), Li & Ramakrishnan (2026), Zhuang et al. (2026), Wang et al. (2026f), Wei et al. (2026b), Rishav et al. (2026), Xu et al. (2026c), Gallego (2026), Zhu et al. (2026b), Jin et al. (2025) ICL-based feedback: Wainrib et al. (2026) Configurations: Yuan et al. (2025b), Kumar et al. (2026c), Gong et al. (2026), Brookes et al. (2025), Huang et al. (2026a), Huang et al. (2026d), Tian (2026), Xie et al. (2026a) Code-level: Hu et al. (2025a), Robeyns et al. (2025), Zhang et al. (2025e), Lee et al. (2026), Pan et al. (2026), Zhang (2026), Zhang et al. (2026o), Zhou et al. (2025c) Multi-agent structural: Yao et al. (2026), He et al. (2026a), Zhang et al. (2026m) Tool-schema optimization: Ghoshal et al. (2026)	Lin et al. (2024)
$\nabla, \mathcal{H}=0$ (Parametric, Autonomous)	$\nabla, \mathcal{H}\neq 0$ (Parametric, Human-involved)
Self-generated signal: Zelikman et al. (2022), Yuan et al. (2024), Wang et al. (2025d), Zhang et al. (2025i), Bhaskar et al. (2025), Khatri et al. (2025), Hübotter et al. (2025), Huang et al. (2026c), Liu et al. (2026f), Lu et al. (2026b), Qu et al. (2026b), Song et al. (2026), Fang et al. (2025d) Multi-role self-play: Chen et al. (2025f), Huang et al. (2025), Peng et al. (2026b), Li et al. (2026i), Li et al. (2026b), Jiang et al. (2026a), Chen et al. (2025a), Liu et al. (2026a), Sui & Hooi (2026), Xue et al. (2025), Wang et al. (2026d), Yuan et al. (2025a), Wei et al. (2025b), Zhao et al. (2025), Xia et al. (2025), Yue et al. (2026), Zhang et al. (2026f), Wu et al. (2025b), Zhang et al. (2026l) Adversarial auditing: Beigi et al. (2026) Self-training: Ni et al. (2024), Liu et al. (2025), Luo et al. (2025), Chen et al. (2026b), He et al. (2025a), Atreja (2025) Tool-integrated: Dong et al. (2025), Feng et al. (2025a), Feng et al. (2026c), Jiang et al. (2025b), Qian et al. (2025), Qin et al. (2024), Li et al. (2025a), Gou et al. (2024), Shang et al. (2025), Cheng et al. (2026a), Zhang et al. (2025h), Zhang et al. (2025j), Zeng et al. (2025), Zhang et al. (2025k), Li et al. (2026g), Kang et al. (2026), Dou et al. (2024) Interactive environments: Chae et al. (2026), Fang et al. (2025c), Qi et al. (2025), Lin et al. (2026d), Dong et al. (2026a), Chen et al. (2025c), Qin et al. (2026), Cai et al. (2026), Liu et al. (2026c) Multi-turn RL methodology: Wang & Ammanabrolu (2025), Wang et al. (2025f), Wei et al. (2025c), Xi et al. (2024), Xi et al. (2025b), Wang et al. (2026h), Zhai et al. (2026), Zou et al. (2026a), Wu & Tang (2026), Yu et al. (2025b), Zhou et al. (2025d), Yang et al. (2026d) World models: Ha & Schmidhuber (2018), Hafner et al. (2025), Feng et al. (2025b), Qiu et al. (2026), Chen et al. (2025g) Continual/personalized: Kim & Kim (2026), Jiang et al. (2025a), Nie et al. (2026b), Hu et al. (2026), Chen et al. (2025e) Multi-LLM collaborative: Jeon et al. (2026), Sharma & Goldwasser (2025), Li & Song (2025), Li et al. (2024c) Meta-learning: Tandon et al. (2025), Wang et al. (2025e), Zweiger et al. (2025) Neuroevolutionary hybrid: Dai et al. (2026), Wang et al. (2025a) Domain-specific: Open Ended Learning Team et al. (2021), Zhang et al. (2024a), Zhang et al. (2026q), Zhang et al. (2026h), Wang et al. (2026g), Zou et al. (2026b), Ye et al. (2025), Yang et al. (2025a), Yin et al. (2024), Black et al. (2025), Sun et al. (2025a), Xiao et al. (2026)	Abramson et al. (2022), Jain et al. (2015), Shivaswamy & Joachims (2015), MacGlashan et al. (2017), Li et al. (2024b), Nam et al. (2026), Yang et al. (2025b), Atreja et al. (2026)
$\Delta+\nabla, \mathcal{H}=0$ (Hybrid, Autonomous)	$\Delta+\nabla, \mathcal{H}\neq 0$ (Hybrid, Human-involved)
Experience consolidation: Qiao et al. (2026), Qian et al. (2026), Shi et al. (2026), Shi et al. (2025), Liao et al. (2026), Ye et al. (2026) Jointly evolving auxiliary structures: Wang et al. (2025b), Wu et al. (2025a), Xia et al. (2026), Wang & Jiang (2026b), Zhai et al. (2025), Yang et al. (2026a), Zhang et al. (2025a), Zhang et al. (2026p), Zhou et al. (2025a), Zhang et al. (2026g), Yu et al. (2026), Ouyang et al. (2026a), Ali et al. (2026), Banerjee et al. (2026), Nie et al. (2026c), Huang et al. (2026b), Yu et al. (2025a), Yuan et al. (2026) Memory/planning joint evolution: Yan et al. (2025), Zhang et al. (2026i), Yang et al. (2026c), Wu et al. (2026a), S1-NexusAgent Team (2026), Zhang et al. (2026j)	—

Self-generated signal: from bootstrapping to verification-driven convergence. Parametric Cortex evolution eliminates external supervision, deriving all training signal from the model’s own generations. Zelikman et al. (2022) establish the generate-filter-train template with STaR; Yuan et al. (2024) extend it to

self-rewarding, where a single LLM serves as both policy and reward model via iterative DPO on self-scored preference pairs. When the evaluator is the model itself (LLM-as-judge) rather than a deterministic verifier, the training signal can still drive initial improvement, but the judge drifts with the system it evaluates, introducing a fragility that deterministic verification avoids. Wang et al. (2025d) formalize the resulting failure mode, *preference signal collapse* (chosen and rejected responses converge, shrinking the score gap by $9\times$ over four iterations), and show that temporal decoupling maintains signal strength. Zhao et al. (2025) demonstrate the opposite end of the spectrum: Absolute Zero Reasoner trains a single LLM to propose and solve code-based reasoning tasks validated solely by a Python executor, achieving state-of-the-art on AIME 2024, LiveCodeBench, and Codeforces without any external dataset. The contrast is sharp: when verification is deterministic, self-generated training signal suffices for sustained improvement; when it is self-referential, signal quality degrades with iteration. The data-free principle generalizes across task domains (Yue et al., 2026; Xia et al., 2025; Wei et al., 2025b; Li et al., 2026i). Multi-role variants distribute the training signal across jointly evolving roles (Chen et al., 2025f; Peng et al., 2026b; Huang et al., 2025; Li et al., 2026b; Liu et al., 2026a; Jiang et al., 2026a; Wang et al., 2026d; Yuan et al., 2025a; Zhang et al., 2026f; Xue et al., 2025; Zhang et al., 2026l; Chen et al., 2025a; Qu et al., 2026b); asymmetric multi-LLM collaboration (Sharma & Goldwasser, 2025; Jeon et al., 2026); adversarial reward auditing (Beigi et al., 2026); bidirectional co-adaptation (Li & Song, 2025); extensions span multimodal domains (Liu et al., 2025; Luo et al., 2025), capability-frontier synthesis (Chen et al., 2026b), self-play RL under data scarcity (Fang et al., 2025d), skill-gap-targeted data (He et al., 2025a), autonomous data generation (Zweiger et al., 2025; Ni et al., 2024; Atreja, 2025; Wang et al., 2025e; Zhang et al., 2025i), test-time adaptation (Tandon et al., 2025; Hübotter et al., 2025), and mid-training reweighting (Huang et al., 2026c) (Table 2).

Environment-grounded training and its instabilities. When the agent operates in stateful environments, parametric training must contend with multi-turn trajectories, sparse delayed rewards, and credit assignment across dozens of actions. In tool-integrated reasoning, Feng et al. (2025a) demonstrate that a 32B model trained to interleave reasoning with code execution surpasses o1-preview on AIME 2024; Li et al. (2026g) apply RL to competitive programming, producing the first AI system to defeat all human competitors in live Codeforces contests. The broader literature converges on cold-start SFT followed by iterative training (Gou et al., 2024; Qin et al., 2024; Dong et al., 2025; Li et al., 2025a; Shang et al., 2025), though some systems train directly from instruction-tuned models via RL (Cheng et al., 2026a; Zhang et al., 2025k), and diverges along multiple design axes—reward granularity (Qian et al., 2025; Zhang et al., 2025h; Feng et al., 2026c), training efficiency (Zhang et al., 2025j; Zeng et al., 2025), capability decomposition (Kang et al., 2026), and framework unification (Jiang et al., 2025b). In interactive digital environments, Qi et al. (2025) construct a self-evolving curriculum that transforms Llama-3.1-8B into a web navigator surpassing GPT-4-Turbo; further work addresses credit decomposition and environment diversity (Dong et al., 2026a; Lin et al., 2026d; Cai et al., 2026; Qin et al., 2026; Wang et al., 2026g; Liu et al., 2026f; Chae et al., 2026; Fang et al., 2025c; Chen et al., 2025c; Wei et al., 2025c; Xi et al., 2024; 2025b) (Table 2). Four failure modes, each identified independently, demonstrate that multi-turn agentic RL introduces qualitatively new instabilities absent from single-turn settings. Wang et al. (2025f) identify the *Echo Trap* (reward variability collapses across batches); Wang et al. (2026h) diagnose *Template Collapse* (training homogenizes strategies into input-agnostic templates, a failure invisible to standard entropy metrics); Zou et al. (2026a) formalize *Information Self-Locking*; and Wu & Tang (2026) identify a three-phase *Reward Hacking Rebound* in code generation. Methodological responses to these instabilities remain fragmented (Wang & Ammanabrolu, 2025), and diagnostic analysis suggests the severity is task-dependent (Zhai et al., 2026). Beyond multi-turn RL, parametric Cortex evolution extends wherever execution-grounded feedback is available: world models learn compressed environment dynamics through imagined rollouts (Ha & Schmidhuber, 2018; Hafner et al., 2025; Feng et al., 2025b) or self-improving training from unlabelled sequences (Qiu et al., 2026); reasoning-based experience synthesis replaces real-environment rollouts with textual simulation (Chen et al., 2025g); lightweight selective adaptation (LoRA with on-policy RL (Hu et al., 2026), likelihood-ratio-triggered updates (Kim & Kim, 2026), federated averaging (Chen et al., 2025e)) outperforms heavyweight continual-learning strategies, and domain-specific applications span open-ended environments (Open Ended Learning Team et al., 2021), embodied control (Black et al., 2025), and further settings (Zou et al., 2026b; Zhang et al., 2024a; 2026q; Li et al., 2024c; Sun et al., 2025a; Xiao et al., 2026; Ye et al., 2025; Yang et al., 2025a; Yin et al., 2024; Yang et al., 2026d) (Table 2). The $\mathcal{H} \neq 0$ papers span sub-optimal corrections (Shivaswamy & Joachims,

2015; Jain et al., 2015) and interactive shaping (MacGlashan et al., 2017), per-user preference learning (Li et al., 2024b; Nam et al., 2026), continual adaptation (Yang et al., 2025b; Abramson et al., 2022), and production-mode deployment (Atreja et al., 2026); Section 5 analyzes these in depth.

Hybrid $\Delta+\nabla$ systems. A growing minority of Cortex papers combine structural and parametric consolidation within a single system. The designs span a temporal spectrum. At one end, *compression lifecycles* treat structural artifacts as scaffolding that is eventually absorbed into weights. Shi et al. (2026) formalize an experience-reflection-consolidation loop: verbal self-reflections generated during RL training are selectively distilled into weights, with the structural reflections discarded once internalized; Ye et al. (2026) instantiate a similar lifecycle for scientific discovery.

At the other end, *persistent joint evolution* maintains both pathways throughout training. Zhou et al. (2025a) provide the cleanest demonstration: INSPO maintains a dynamic population of system instructions that undergo mutation and selection in tandem with policy updates. Zhang et al. (2026g) offer a principled rationale: system prompts encode declarative knowledge (Δ) while weights encode procedural knowledge (∇), and their joint optimization yields strong mathematical reasoning performance. Additional targets of persistent joint evolution include skill libraries (Xia et al., 2026; Wang et al., 2025b; Ouyang et al., 2026a), experience bases (Wu et al., 2025a; Wang & Jiang, 2026b; Zhang et al., 2026p), trajectory summaries (Zhai et al., 2025), filtered rollout pools (Zhang et al., 2025a), weakness-driven task synthesis (Yang et al., 2026a), and learnable soft tokens (Yu et al., 2026). When the jointly evolving target is external memory, the agent must learn when and how to manage its own knowledge store: Yan et al. (2025) train memory management operations via RL using downstream answer correctness as the sole reward, and further systems jointly evolve persona memory (Zhang et al., 2026i), skill banks (Wu et al., 2026a), and jointly evolving memory with parametric components (Qiao et al., 2026; Qian et al., 2026; Liao et al., 2026; Shi et al., 2025; Yang et al., 2026c; S1-NexusAgent Team, 2026; Zhang et al., 2026j) (Table 2). Banerjee et al. (2026) demonstrate that hybrid joint evolution can enforce formal correctness: their Formally Guarded Generative Model couples Dafny-based verification (Δ on $\mathcal{A}$) with RL fine-tuning (∇ on Π), achieving zero constraint violations; Ali et al. (2026) jointly evolve a security constitution with parametric unlearning; further hybrids include Yuan et al. (2026). At the boundary between structural and parametric, neuroevolutionary hybrids (Dai et al., 2026; Wang et al., 2025a; Su et al., 2026a) occupy an intermediate position that resists clean classification: each combines gradient-free evolution of latent embeddings or code populations with gradient-based policy training, so neither the structural nor the parametric label fully applies. Section 4 analyzes the design trade-offs between compression and persistent joint evolution across all three substrates.

Synthesis: what governs Cortex evolution. The Cortex results with the strongest absolute benchmark performance all rely on deterministically verifiable evaluation signals: AZR’s code executor (Zhao et al., 2025), GrandCode’s contest judge (Li et al., 2026g), ReTool’s test suite (Feng et al., 2025a). Where such verification is absent, iterative improvement degrades through distinct mechanisms: preference signal collapse when the evaluator shares parameters with the system it evaluates (Wang et al., 2025d), and template collapse when multi-turn optimization homogenizes strategies beyond what standard metrics detect (Wang et al., 2026h). Domains lacking deterministic verification remain largely out of reach, though initial steps target non-verifiable reasoning (Sui & Hooi, 2026), general-purpose chat (Bhaskar et al., 2025), and multiturn collaborative tasks (Wu et al., 2025b). Whether this regularity holds beyond Cortex is a question Section 5 takes up across all three substrates.

Independent of evaluation quality, regression emerges as a recurring concern. Five independent systems converge on gated-admission mechanisms that prevent iterative improvement from erasing prior gains (Zhang, 2026; Gong et al., 2026; Zhang et al., 2026k; Zhou et al., 2025a; Banerjee et al., 2026); Section 4 examines whether this pattern generalizes beyond Cortex.

Several gaps limit maturity: long-term evolution dynamics (most papers evaluate fewer than ten iterations), safety constraints (only 5 of nearly 200 papers (Banerjee et al., 2026; Ali et al., 2026; Beigi et al., 2026; Zhang et al., 2026f; Gallego, 2026)), sparse analytical or scaling-law work (Jeon et al., 2026; Khatri et al., 2025; Banerjee et al., 2026), limited cross-domain transfer testing (Liu et al., 2026a; Cheng et al., 2026a; Xi et al., 2024), and rarely reported computational costs.

Cortex evolution changes *how* the agent reasons; Action evolution determines the interface through which deliberation translates into environmental effect.

Key Takeaways: Cortex

- **Evaluation-signal verifiability predicts success.** The Cortex results with the strongest benchmark performance all rely on deterministically verifiable evaluation signals (code execution, exact match, game outcomes); where such signals are absent, iterative improvement degrades. Section 5 examines whether this regularity holds across substrates.
- **Hybrid $\Delta+\nabla$ joint evolution is a growing pattern.** A substantial minority couple structural artifacts with parametric weight updates; system prompts encode declarative knowledge (Δ) while weights encode procedural knowledge (∇). Section 4 analyzes the pathway-level regularities.
- **Regression avoidance is a recurring concern.** Five independent systems converge on gated-admission mechanisms; Section 4 examines whether this concern tracks consolidation pathway rather than substrate.
- **Safety remains a critical gap.** Only 5 of nearly 200 Cortex papers constrain autonomous evolution with safety or alignment mechanisms.

3.2 Action: Evolution of the Executive Substrate

Action ($\mathcal{A}$) is the executive substrate (toolset, execution logic, multi-agent orchestration), accounting for about 75 of the surveyed evolution papers. Unlike Cortex, where the $\Delta-\nabla$ continuum structures the literature, Action evolution is overwhelmingly structural. The evolved artifacts range from atomic tools through multi-step skills and workflow topologies to complete agent codebases; expressive power increases along this range but evaluability decreases. Table 3 provides the complete paper listing.

Skill accumulation is easy; curation is the bottleneck. The *library-building* paradigm is the most common approach, sharing a lifecycle (creation $\rightarrow$ storage $\rightarrow$ retrieval $\rightarrow$ maintenance) applicable to both atomic tools and multi-step skills. Tool-creation methods derive tools from execution traces (Abuzakuk et al., 2026), reference materials (Liu et al., 2026g), LLM-generated code (Cheng et al., 2026b), scratch generation (Li et al., 2026d), external repositories (Jin et al., 2025), API refinement (Qu et al., 2025), task-driven tool generation (Zhang et al., 2026t), and human expert feedback (Gao et al., 2025b); Feng et al. (2026c) jointly optimize tool creation with an RL-trained policy ($\Delta+\nabla$). Wang et al. (2024a) establish the skill accumulation paradigm in Minecraft: an LLM agent iteratively generates, verifies, and stores JavaScript skill code, producing an ever-expanding behavioral repertoire with no mechanism for pruning. The field’s response spans a curation spectrum: from accumulation without explicit curation (Qiu et al., 2025b) through failure-attributed skill rewriting with unit-test gates (Zhou et al., 2026) and five explicit maintenance operations (Wu et al., 2026a) to dedicated RL-trained curators (Ouyang et al., 2026a). Cross-user skill aggregation (Ma et al., 2026b), user-feedback-driven maturity tracking (Zhang et al., 2026a), and progressive dialogue-driven service generation (Adnan et al., 2025) account for three of the four $\mathcal{H} \neq 0$ designs in Action (the fourth, Gao et al. 2025b, appears above); further work spans experience-to-tool conversion (Shen et al., 2026c), subagent packaging (Zhang et al., 2026s), surrogate verifiers (Zhang et al., 2026c), scientific repository mining (Shen et al., 2026a), failure analysis (Liu et al., 2026h), complementary skill families (Jiang et al., 2026c), and skill consolidation (Wang et al., 2026a; Qiu et al., 2025a; Liang et al., 2026) (Table 3).

Population-based search with deterministic evaluation. In evolutionary program search, the LLM produces offspring that a deterministic evaluator scores. Romera-Paredes et al. (2024) pair evolutionary search with deterministic evaluation, producing the first LLM-driven scientific discoveries on established open problems (new cap-set constructions, improved bin-packing heuristics). Novikov et al. (2025) scale the paradigm to full-file evolution with evaluation cascades, yielding the first improvement to Strassen’s 4×4 complex matrix multiplication algorithm in 56 years. Banerjee et al. (2026) achieve the strongest formal guarantee in this space: evolutionary search paired with formal Dafny verification ensures provable correctness alongside RL training, a cross-substrate hybrid (Δ on $\mathcal{A}$, ∇ on Π). Further architectures span autonomous kernel optimization (Chen et al., 2026a), quality-diversity archives (Lehman et al., 2022; Kumar

et al., 2026a), adaptive model routing for cost efficiency (Ray et al., 2026), evaluation-driven discovery (Ye et al., 2026), verbal gradient feedback (Gungordu et al., 2026), and asynchronous multi-step agents (Hambardzumyan et al., 2026) (Table 3).

Topology and whole-system evolution. Evolving the agent’s execution topology trades expressiveness against searchability: Hu et al. (2025a) represent agents as arbitrary Python code (Turing-complete); Zhang et al. (2026m) constrain evolution to atomic FSM operations (interpretable, reversible); Zhang et al. (2025f) use MCTS over reusable operators. Further designs span symbolic gradients (Zhou et al., 2025c), DAG-based hierarchies (Huang et al., 2026a; Xu et al., 2026a; Yu et al., 2025a), natural language programs (Li et al., 2024c), Python harness programs (Lee et al., 2026), and observability-driven approaches with auto-reversion (Lin et al., 2026a) (Table 3). At the coarsest level, whole-system evolution modifies the agent’s entire codebase: Weng et al. (2026) show that group evolution outperforms individual evolution (Zhang et al., 2025e) on SWE-bench Verified; Zhang (2026) apply stability-prioritizing gating; Robeyns et al. (2025) eliminate the meta/object-agent distinction; Wang et al. (2026f) jointly evolve cognition, actions, and memory. Domain-specific variants provide evaluation signals independent of the agent being evolved through formal verification (Yu & Ren, 2026), production A/B testing (Wang et al., 2026b), and further approaches (Xu et al., 2026b; Liu et al., 2026e; Xiong et al., 2026) (Table 3).

Synthesis. The same regularity visible in Cortex appears here: the strongest Action results co-occur with the strongest evaluation signals (formal verification (Banerjee et al., 2026), execution-based scoring (Romera-Paredes et al., 2024; Novikov et al., 2025), production A/B testing (Wang et al., 2026b)), regardless of search sophistication. Because execution-based evaluation already provides reliable signal without gradient access, parametric methods remain confined to narrow niches: purely parametric Action papers learn routing or communication models (Jiang et al., 2026b; Ong et al., 2025); two further routing systems combine parametric training with structural artifact evolution (Huang et al., 2026b; Nie et al., 2026c). Configuration optimization treats the agent pipeline as a tunable system without weight updates (Brookes et al., 2025; Ghoshal et al., 2026; Guan et al., 2026); program-space search evolves executable strategies (Gautam et al., 2026); Li et al. (2025b) offer a decentralized structural alternative. Code is the predominant representation across all granularity levels (Romera-Paredes et al., 2024; Wang et al., 2024a; Hu et al., 2025a; Zhang et al., 2025e; Weng et al., 2026), reinforcing structural dominance. Cross-domain transfer and long-horizon evolution remain untested (Xiong et al., 2026).

Key Takeaways: Action

- **Evaluation quality governs outcomes.** The same regularity visible in Cortex appears in Action: the strongest results co-occur with the strongest evaluation signals, regardless of search sophistication. Section 5 examines this cross-substrate pattern.
- **Code is the predominant representation.** Executable code serves as evolved artifact, mutation medium, and evaluation substrate across most of the hierarchy, reinforcing the overwhelming structural dominance.
- **Near-absence of human selective pressure.** Execution feedback provides objective success signals, reducing the need for human judgment that is more prominent in Cortex and Memory.

3.3 Memory & Sense: Evolution of the Epistemic Substrate

Of the three substrates, Memory & Sense ($\mathcal{M}$) exhibits extreme structural dominance comparable to Action: nearly all of about 100 papers use structural consolidation (Δ). We organize this subsection along the *memory lifecycle*, from accumulation through curation to meta-evolution of the architecture itself, because the Δ - ∇ continuum that organizes Cortex would place nearly all papers in a single category. Table 4 provides the complete paper listing.

Abstraction level determines transferability. The dominant form follows a shared pipeline (*execute* $\rightarrow$ *extract* $\rightarrow$ *store* $\rightarrow$ *retrieve* $\rightarrow$ *reuse*), differentiated by abstraction level, feedback signal, and failure handling. Memory entries range from raw trajectories (Fang et al., 2025b; Sarukkai et al., 2025) through

Table 3: **Surveyed paper listing for Action evolution** (about 75 papers), organized by consolidation pathway (Δ vs. ∇) and selective pressure ($\mathcal{H}=0$ vs. $\mathcal{H}\neq 0$). Hybrid labels are system-level. Sub-group labels are approximate; individual papers may span multiple categories.

$\Delta, \mathcal{H}=0$ (Structural, Autonomous)	$\Delta, \mathcal{H}\neq 0$ (Structural, Human-involved)
Tools & Skills: Abuzakuk et al. (2026), Cheng et al. (2026b), Jin et al. (2025), Li et al. (2026d), Lin et al. (2026b), Liu et al. (2026g), Qu et al. (2025), Jiang et al. (2026c), Liang et al. (2026), Liu et al. (2026h), Zhou et al. (2026), Qiu et al. (2025b), Qiu et al. (2025a), Shen et al. (2026a), Shen et al. (2026c), Wang et al. (2024a), Wang et al. (2026a), Zhang et al. (2026t), Zhang et al. (2026s), Zhang et al. (2026c) Program Search: Chen et al. (2026a), Gungordu et al. (2026), Hambardzumyan et al. (2026), Kumar et al. (2026a), Lehman et al. (2022), Liu et al. (2026e), Novikov et al. (2025), Ray et al. (2026), Romera-Paredes et al. (2024), Ye et al. (2026), Zhang et al. (2025b), Gautam et al. (2026) Workflow & Orchestration: He et al. (2026a), Hu et al. (2025a), Huang et al. (2026a), Lee et al. (2026), Li et al. (2025b), Lin et al. (2026a), Li & Ramakrishnan (2026), Xu et al. (2026a), Zhang et al. (2025f), Zhang et al. (2026o), Zhang et al. (2026m), Zhou et al. (2025c) Whole-System: Robeyns et al. (2025), Wang et al. (2026f), Weng et al. (2026), Xu et al. (2026b), Xiong et al. (2026), Yu & Ren (2026), Zhang et al. (2025e), Zhang (2026), Wang et al. (2026b) Configuration: Brookes et al. (2025), Ghoshal et al. (2026), Guan et al. (2026) Cross-cutting: Qu et al. (2026a), Nie et al. (2026a)	Tools & Skills: Adnan et al. (2025), Gao et al. (2025b), Ma et al. (2026b), Zhang et al. (2026a)
$\nabla, \mathcal{H}=0$ (Parametric, Autonomous)	$\nabla, \mathcal{H}\neq 0$ (Parametric, Human-involved)
Black et al. (2025), Jiang et al. (2026b), Ong et al. (2025)	—
$\Delta+\nabla, \mathcal{H}=0$ (Hybrid, Autonomous)	$\Delta+\nabla, \mathcal{H}\neq 0$ (Hybrid, Human-involved)
Banerjee et al. (2026), Dong et al. (2026a), Feng et al. (2026c), Huang et al. (2026b), Li et al. (2024c), Nie et al. (2026c), Su et al. (2026a), Wang et al. (2025b), Wu et al. (2026a), Ouyang et al. (2026a), Yu et al. (2025a)	—

workflows (Wang et al., 2025g) to highly abstracted insights (Cai et al., 2025b; Ouyang et al., 2026b). Kim et al. (2026) provide the clearest evidence that abstraction dictates transferability: insights transfer better than summaries, summaries better than workflows, and workflows better than raw trajectories. Further systems populate the spectrum with intermediate artifacts (Bai et al., 2025; Zhuang et al., 2024; Li et al., 2026f), versioned registries (Lin et al., 2026b), dynamically maintained knowledge bases (Cai et al., 2025a), and domain-tailored schemas that achieve precise within-domain retrieval at the cost of portability (Ren et al., 2026b; Shen et al., 2026b; Zhang et al., 2026n; Zhu et al., 2025) (Table 4).

Multiple independent papers demonstrate that failures produce informative memory entries (Xie et al., 2026b; Ren et al., 2026b; Yu et al., 2026; Wang & Jiang, 2026a); yet most systems store only successful outcomes, leaving what Ren et al. (2026b) call the error-driven dividend unrealized. Feedback signal reliability varies from ground-truth comparison (Chen et al., 2025b; Li et al., 2024a) through LLM-as-Judge (Ouyang et al., 2026b; Cao et al., 2025) to domain tool execution (SPICE simulation (Bao et al., 2026), GPU profiling (Dong et al., 2026b), EDA synthesis (Fang et al., 2026)), with multi-stage verification providing stronger quality controls (Li et al., 2026e; Liao et al., 2026). Embodied (Xie et al., 2026b; Wei et al., 2026a; Wang et al., 2026c; Li et al., 2026a), game-strategic (Guan et al., 2024; Ma et al., 2026a), research-oriented (Lyu et al., 2026; Long, 2026), and multi-agent settings with per-agent memory and emergent cooperation (Xie, 2026; Qu et al., 2026a; Liu et al., 2026d) share the same accumulation-versus-curation trade-offs (Table 4).

Persistent relationships favor graph-based representations. When memory encodes persistent relationships, graph-based representations dominate (Zhu et al., 2026a; Rasmussen et al., 2025; Zheng et al., 2026; Han et al., 2026; Zhang et al., 2026e; 2025c; Wu et al., 2026b; Chhikara et al., 2025); Lu et al. (2026a) show that the optimal topology varies with context, with a learned classifier selecting among linear, graph, and hierarchical structures per episodic unit; structured-schema alternatives without explicit graph topology also persist (Gadzhiev & Kislov, 2026; Packer et al., 2023; Xu et al., 2025a; Westhäußer et al., 2025); narrative simulation presents a distinct variant (He et al., 2026b) (Table 4). The stability-plasticity dilemma (superseding outdated beliefs without destroying useful context) is addressed by renormalization group theory (Tian et al., 2025), Piaget-inspired assimilation and accommodation (Zheng et al., 2026), and Hebbian decay (Zhu et al., 2026a), though explicit forgetting remains rare.

Fang et al. (2025b) provide the most systematic study of the accumulation ceiling. Their scaling analysis shows that retrieval count exhibits an optimal range, beyond which excess memories degrade performance. Cao et al. (2025) demonstrate the most complete lifecycle, showing that memory quality can substitute for

model scale; to our knowledge, the only Memory-substrate system where a smaller model with evolved memory outperforms a larger memoryless model. Additional lifecycle innovations include Meta-MDP operations (Cai et al., 2025b), test-time logit modulation (Li et al., 2026h), and live memory evolution (Zhang et al., 2026r).

He et al. (2025b) report the largest-scale deployed $\mathcal{H} \neq 0$ memory system in the survey, deployed at TikTok Pay across 150M+ monthly active users: the system assesses its own uncertainty and queries human experts when confidence is moderate or low, achieving high sensitivity in production. This result illustrates the role of human selective pressure: when domains involve non-stationary rules and value-laden judgment, autonomous signals are insufficient. Additional human-involved systems include Deng et al. (2026); Mozannar et al. (2025); Chen et al. (2026c); Zhang et al. (2026a) (Table 4).

Curation outperforms accumulation but adds complexity. When entries carry quality scores, population-based operators yield some of the strongest purely structural results. Wang & Jiang (2026a) propose PRIME with biologically inspired curation operators (mutation, generalization, crossover, pruning). Wu et al. (2025a) couple dynamic scoring with RL policy training ($\Delta + \nabla$). Further curation systems include constrained FSM operators (Zhang et al., 2026m), RL-assigned quality scores (Wang & Jiang, 2026b; Xia et al., 2026), multi-agent memory coordination (Lin et al., 2026c), and bounded optimization memories (Zhang et al., 2025b). Retrieval innovation evolves the form of retrieved artifacts: query-conditioned reasoning abstractions (Feng et al., 2026b), meta-buffers of reasoning templates (Yang et al., 2024b), graph-based evidence structures (Penarozza, 2026), and evolving latent-preference hypotheses (Yoon et al., 2026). Formal frameworks unifying curation designs include the Stateful Reflective Decision Process (Wang, 2025) and the Memory-augmented MDP (Zhou et al., 2025b); Lam et al. (2026)’s SSGM provides reconciliation-bounded governance for memory safety; none accounts for evolutionary curation operators.

Forgetting: critically understudied. Despite evidence that unbounded accumulation degrades performance (Fang et al., 2025b; Kwon et al., 2026), only fewer than ten of about 100 papers implement explicit forgetting. ZenBrain (Bering, 2026) deploys multiple neuroscience-inspired mechanisms but reveals non-additive interactions limiting adoptability. Memory Worth (Simsek, 2026) offers a simpler two-counter signal with convergence guarantees. Between these extremes: Ebbinghaus-inspired decay with adaptive reinforcement (Gu et al., 2026). An outlier, TACO (Ren et al., 2026a), evolves terminal output compression rules, the only paper evolving the Sense component of $\mathcal{M}$.

Meta-evolution and hybrid designs. A small but growing body of work searches over memory architectures themselves: MSTAR (Pan et al., 2026) evolves executable Python programs defining memory schemas; MemEvolve (Zhang et al., 2025d) evolves the full pipeline via multi-objective selection; Zhang et al. (2026d) reframe operations as learnable skills with an RL-trained controller ($\Delta + \nabla$); and PRISM (Mishra, 2026) unifies memory types under a decision-theoretic framework with convergence guarantees. Only 4 of about 100 papers consolidate purely parametrically: world models (Hafner et al., 2025; Ha & Schmidhuber, 2018), test-time training (Sun et al., 2025b), and selective sparse memory updates (Lin et al., 2025). An emerging hybrid pattern, *store structurally, retrieve parametrically*, appears independently across multiple systems (Zhou et al., 2025b; Zhang et al., 2026d; Wu et al., 2025a; Wang & Jiang, 2026b; Cai et al., 2026; Kim & Kim, 2026; Liao et al., 2026; Qiao et al., 2026; Zhang et al., 2026j; Zhai et al., 2025; Zhang et al., 2026p; S1-NexusAgent Team, 2026; Yang et al., 2026c) (Table 4).

Synthesis. The accumulation ceiling documented above echoes the context-overflow problem in structural Cortex (§3.1): in Cortex, accumulated artifacts exceed the context window’s capacity; in Memory, retrieval precision degrades as the collection grows. The bottleneck differs (capacity in Cortex, precision in Memory) but the shared root is that structural artifacts grow with experience while no compression mechanism reduces them. Section 4 treats this as a pathway-level regularity. The “store structurally, retrieve parametrically” hybrid that appears independently across 13 systems (§3.3) is a pattern without clear counterpart in Cortex or Action, where hybrids typically follow a compression or persistent-joint-evolution lifecycle. Forgetting remains critically understudied (fewer than ten of about 100 papers), and Sense evolution is represented by a single paper (Ren et al., 2026a). Individual papers borrow from cognitive science (Ebbinghaus forgetting (Zhong et al., 2024; Gu et al., 2026), Hebbian plasticity (Zhu et al., 2026a), Piaget’s schemas (Zheng et al.,

2026), free-energy principle (Ma et al., 2025), sleep-based consolidation (Bering, 2026)), but no paper in the surveyed corpus maps the full space of cognitive memory mechanisms onto agent memory design.

Key Takeaways: Memory & Sense

- **The accumulation ceiling parallels Cortex context overflow.** In Cortex, artifacts exceed context capacity; in Memory, retrieval precision degrades as the collection grows. The shared root is that structural artifacts accumulate without compression; Section 4 treats this as a pathway-level regularity.
- **Extreme structural dominance with an emerging hybrid pattern.** Nearly all papers use structural consolidation (Δ). Thirteen papers independently converge on a “store structurally, retrieve parametrically” hybrid without clear counterpart in Cortex or Action.
- **Failures are more informative than successes, yet underexploited.** Four independent papers document this error-driven dividend across multiple settings, yet most systems store only successful outcomes.
- **Forgetting and Sense evolution are critically understudied.** Only ~ 8 of about 100 papers implement explicit forgetting; Sense evolution is represented by a single paper (Ren et al., 2026a).

Table 4: **Surveyed paper listing for Memory & Sense evolution** (about 100 papers), organized by consolidation pathway and selective pressure. Hybrid labels are system-level. Sub-group labels are approximate; individual papers may span multiple categories.

$\Delta, \mathcal{H}=0$ (Structural, Autonomous)	$\Delta, \mathcal{H}\neq 0$ (Structural, Human-involved)
Accumulation:Task-Oriented: Bai et al. (2025), Bao et al. (2026), Cai et al. (2025a), Cai et al. (2025b), Cao et al. (2025), Chen et al. (2025b), Cheng et al. (2026b), Dong et al. (2026b), Fang et al. (2025b), Fang et al. (2026), Gautam et al. (2026), Jiang et al. (2026c), Kim et al. (2026), Li et al. (2024a), Li et al. (2026a), Li & Ramakrishnan (2026), Li et al. (2026b), Li et al. (2026f), Li et al. (2026e), Liang et al. (2026), Lin et al. (2026b), Lyu et al. (2026), Ouyang et al. (2026b), Sarukkai et al. (2025), Wang et al. (2025g), Zhuang et al. (2024) Accumulation:Embodied: Wang et al. (2026c), Wei et al. (2026a), Xie et al. (2026b) Accumulation:Game-Strategic: Guan et al. (2024), Ma et al. (2026a), Xie et al. (2026a) Accumulation:Domain-Specific: Han et al. (2026), Liu et al. (2026d), Ren et al. (2026b), Shen et al. (2026b), Zhang et al. (2026n), Zhu et al. (2025) Accumulation:Multi-Agent: Qu et al. (2026a), Wang et al. (2026f), Xie (2026), Zhang et al. (2025c) Accumulation:Research/Discovery: Liu et al. (2026e), Long (2026), Xu et al. (2026b) Accumulation:Conversational/Persona: He et al. (2026b), Kwon et al. (2026), Lu et al. (2026a), Ma et al. (2025), Park et al. (2023), Wu et al. (2026b), Xu et al. (2025a), Zhang et al. (2026e), Zheng et al. (2026), Zhu et al. (2026a) Curation/Operators: Lin et al. (2026c), Wang & Jiang (2026a), Zhang et al. (2025b), Zhang et al. (2026r), Zhang et al. (2026m) Forgetting/Lifecycle: Bering (2026), Gu et al. (2026), Simsek (2026), Xu et al. (2026c) Meta-Design: Mishra (2026), Pan et al. (2026), Zhang et al. (2025d) Retrieval Innovation: Feng et al. (2026b), Penarozza (2026), Yang et al. (2024b), Yoon et al. (2026) Sense/Compression: Ren et al. (2026a) Benchmark: Wei et al. (2025a) Theoretical: Wang (2025)	Conversational/Persona: Chhikara et al. (2025), Gadzhiev & Kislov (2026), Nie et al. (2026b), Packer et al. (2023), Rasmussen et al. (2025), Tian et al. (2025), Westhäüßer et al. (2025), Zhong et al. (2024) Human-in-the-Loop: Chen et al. (2026c), Deng et al. (2026), He et al. (2025b), Mozannar et al. (2025), Zhang et al. (2026a)
$\nabla, \mathcal{H}=0$ (Parametric, Autonomous)	$\nabla, \mathcal{H}\neq 0$ (Parametric, Human-involved)
Ha & Schmidhuber (2018), Hafner et al. (2025), Sun et al. (2025b), Lin et al. (2025)	—
$\Delta+\nabla, \mathcal{H}=0$ (Hybrid, Autonomous)	$\Delta+\nabla, \mathcal{H}\neq 0$ (Hybrid, Human-involved)
Experience+RL: Qiao et al. (2026), Yu et al. (2026), Zhai et al. (2025), Zhang et al. (2026p) Structural+RL: Cai et al. (2026), Liao et al. (2026), Wang & Jiang (2026b), Wu et al. (2025a), Xia et al. (2026), Zhang et al. (2026d) Domain Hybrid: Feng et al. (2025b), S1-NexusAgent Team (2026), Yang et al. (2026c), Zhang et al. (2026j) Theoretical: Zhou et al. (2025b) Persona: Kim & Kim (2026)	—

3.4 Cross-Substrate Interactions

The preceding three subsections treat Cortex, Action, and Memory as independent evolutionary targets. In practice, the substrates are coupled: changes in one propagate to, constrain, or enable changes in another. Of the about 300 evolution papers, about 70 receive classifications in multiple taxonomy cells.

The Cortex–Action coupling (27 papers) is the most common, partly because RL-based training of tool-use policies provides well-established reward signals. The Cortex–Memory coupling (20 papers) reflects structural injection of retrieved knowledge into context and parametric internalization into weights. Action–Memory coupling (10 papers) is least common because skill libraries are often treated as static rather than jointly evolving. Only three papers close the full $\Pi \rightarrow \mathcal{A} \rightarrow \mathcal{M} \rightarrow \Pi$ loop; HERA (Li & Ramakrishnan, 2026), EvoFSM (Zhang et al., 2026m), and AutoAgent (Wang et al., 2026f) all use structural (Δ) pathways. Zhang et al. (2026o) occupies an intermediate position with a protocol-level three-substrate design but only two-substrate experiments.

Coupling directions. *Cortex* $\rightarrow$ *Action*: improving Cortex tends to improve Action effectiveness, but expanding Action does not improve Cortex unless the deliberative substrate is retrained; the coupling tightens under joint evolution (Feng et al., 2026c; Nie et al., 2026c). *Action* $\rightarrow$ *Memory*: skill libraries straddle the $\mathcal{A}/\mathcal{M}$ boundary, and curation mechanisms from Memory apply directly. *Memory* $\rightarrow$ *Cortex*: takes structural injection (Kim et al., 2026) and parametric internalization (Lu et al., 2026b; Shi et al., 2026) forms; the “store structurally, retrieve parametrically” hybrid (Zhou et al., 2025b; Zhang et al., 2026d) directly manifests this coupling.

System-level implications. The substrates form a directed cycle; systems that close it target the broadest improvements but face the hardest credit assignment. Consolidation pathway is partially determined by coupling direction. No paper in the surveyed corpus studies dynamics of cross-substrate coupling over extended horizons.

Key Takeaways: Cross-Substrate Interactions

- **Substrates form a directed cycle, but full-loop closure is rare.** Only three papers close the full $\Pi \rightarrow \mathcal{A} \rightarrow \mathcal{M} \rightarrow \Pi$ loop, all via Δ pathways.
- **Consolidation pathway is partially determined by coupling direction.** Cortex–Action is pathway-mixed, Action–Memory is predominantly Δ .
- **Cross-substrate dynamics over time remain unstudied.** No paper in the surveyed corpus characterizes whether improvements in one substrate accelerate, saturate, or interfere with evolution in another.

Substrate choice constrains the consolidation pathway: Cortex spans the full Δ – ∇ continuum, while Action and Memory are overwhelmingly structural. Section 4 examines this continuum directly, asking *how* transient experience becomes persistent change.

4 Axis II: Consolidation Pathway

4.1 Constraints on Pathway Selection

Having analyzed what evolves in each substrate (Section 3), we now examine how change persists: the consolidation pathway axis. The structural majority among the evolution papers is not uniformly distributed across substrates; it reflects three reinforcing constraints that jointly determine which pathway a system adopts. The clearest asymmetry is a parametric confinement pattern: only a handful of Memory and Action papers use purely parametric consolidation on their respective substrates, compared with the large majority of Cortex papers. This pattern is predictable from three properties of each substrate, none of which is a free design choice.

Artifact discreteness. Action artifacts (tool specifications, skill code, workflow graphs) and Memory entries (facts, rules, cases) are inherently symbolic; gradients have no natural target in a Python function’s structure or a knowledge-graph triple. Hu et al. (2025a) illustrate the converse: representing agents as code provides Turing-complete expressiveness, and the discrete, symbolic nature of the search space makes LLM-based mutation a natural optimization mechanism. Cortex weights, by contrast, are continuous by construction, and the entire RL and SFT apparatus is designed for differentiable objectives. This discrete-

ness asymmetry explains why parametric methods concentrate in Cortex while remaining confined to narrow niches in Action (orchestration routing and continuous control, only 5 papers) and Memory (world models, test-time training, and sparse memory finetuning, only 4 papers), the only sub-problems within those substrates that present continuous optimization targets in the surveyed literature.

Evaluation signal type. Parametric consolidation achieves its strongest results when rewards are deterministically verifiable (code execution, exact-match answers, game outcomes) because continuous optimization in weight space, whether gradient-based or derivative-free, requires dense, reliable signal. Zhao et al. (2025) demonstrate the limiting case: their data-free self-play system achieves state-of-the-art mathematical and coding reasoning with zero external data, because code executors provide deterministically verifiable rewards at zero annotation cost. Section 5 formalizes this as the strongest tier in a three-tier signal hierarchy. Where such signals are unavailable (open-ended dialogue, knowledge curation, strategy selection), structural methods that rely on LLM-as-judge or heuristic scoring dominate instead. The reward signal hierarchy identified within Cortex (Section 3.1) thus generalizes to a *pathway-level* regularity: verifiable rewards enable parametric consolidation regardless of substrate, while noisy or absent reward signals select for structural methods.

Infrastructure access. When the base model is a frozen proprietary API, structural consolidation is not a preference but a necessity: no gradient signal can reach weights the developer cannot access. Early agentic evolution systems illustrate this constraint directly: Park et al. (2023)’s generative agents and Shinn et al. (2023)’s Reflexion are both purely structural designs requiring no weight updates, a property that made them naturally compatible with the API-only deployment regime that dominated 2023. As open-weight models (LLaMA, Qwen, DeepSeek) proliferate and RL infrastructure matures, the parametric share grows: the GRPO-based training adopted by roughly 40 Cortex papers would be impossible without weight access, and nearly all hybrid papers appear in 2025–2026. This temporal pattern suggests that the current distribution is a snapshot of a shifting frontier rather than a stable equilibrium, with the structural–parametric balance determined partly by infrastructure availability at the time of development.

How the three constraints interact. These three constraints (artifact discreteness, evaluation signal type, and infrastructure access) are not independent: they reinforce one another to produce a single overarching trade-off. Structural methods are more accessible (no training infrastructure, interpretable artifacts, compositional modularity) but hit hard ceilings (finite context, unbounded accumulation, bounded search budgets); parametric methods escape those ceilings through continuous compression into weights but require weight access, verifiable rewards, and risk instabilities (template collapse, reward hacking, catastrophic forgetting) absent from structural evolution. Neither pathway dominates; the optimal choice depends on the substrate’s discreteness, the domain’s reward structure, and the deployment setting’s infrastructure constraints. These three factors jointly explain why purely parametric methods are nearly absent from Action and Memory while prevailing in Cortex, and why the hybrid share remains small but is growing rapidly. The distributional asymmetry, however, is only the surface pattern; underneath it lie different failure landscapes, shared within pathway rather than within substrate, whose structural and parametric manifestations the remainder of this section examines in turn.

4.2 Structural Consolidation

Structural methods across all three substrates share four properties that explain both their dominance and their limits: human-readable artifacts, no training infrastructure requirement, compositional modularity, and a hard ceiling imposed by finite context or finite search budgets. These properties are pathway-level regularities invisible in Section 3’s substrate-by-substrate narrative, where context overflow in Cortex prompts, library growth in Action skills, and retrieval degradation in Memory are discussed as separate, substrate-specific challenges rather than manifestations of a single structural constraint.

Interpretability and compositionality. Whether the evolved artifact is a prompt entry, a skill program, or a memory record, structural consolidation produces outputs that humans can inspect, edit, and selectively roll back, a property parametric methods lack at the level of individual artifacts. Evolved constitutions for

multi-agent coordination (Kumar et al., 2026c), JavaScript skill code in Minecraft (Wang et al., 2024a), and timestamped knowledge entries in a production compliance system (He et al., 2025b) are all auditable without interpretability tools, making structural evolution the most straightforward pathway for high-stakes settings where accountability requires tracing each change to its source. Compositionality compounds this advantage: structural artifacts can be selectively combined, deleted, or version-controlled, enabling the regression-avoidance mechanisms that independently emerge across substrates. Zhang (2026) introduce gated admission for joint Cortex–Action evolution; Zhou et al. (2026) impose test-based gates on Action skill mutations; and Liao et al. (2026) apply confidence-based gating in Memory. That three independent systems converge on gated admission, each in a different substrate, is consistent with regression control being a pathway-level concern rather than a substrate-specific design choice.

Unbounded accumulation. The complementary weakness of compositionality is monotonic growth: structural artifacts accumulate without natural compression, and the resulting growth degrades performance through qualitatively different mechanisms across substrates that share a common root. In Cortex, Reflexion’s bounded reflection buffer sacrifices long-horizon experience for immediate relevance, limiting context overhead from unbounded accumulation. Zhang et al. (2026k) diagnose the underlying failure mode as *context collapse*: monolithic rewriting by LLMs degrades accumulated knowledge into progressively shorter summaries. In Action, Wang et al. (2024a)’s append-only skill library grows without pruning, retrieving only the top-5 most relevant skills per task, a fixed window over a growing collection; Wu et al. (2026a) respond with explicit curation operations governed by trajectory-derived contracts, treating library curation as a first-class design concern rather than an afterthought. In Memory, Fang et al. (2025b)’s scaling analysis identifies an optimal retrieval count beyond which additional entries degrade performance, and Simsek (2026) formalize the deprecation problem through a principled Memory Worth metric with convergence guarantees, specifying when accumulated structural artifacts should be suppressed or removed. The shared root cause is that structural artifacts occupy representational space (context tokens, library indices, retrieval candidates) that scales linearly with experience, while the mechanisms that select among them do not scale accordingly. That six independent systems across three substrates converge on bounded retrieval, active curation, or principled deprecation, each targeting the same linear-growth bottleneck through different mechanisms, is consistent with unbounded accumulation being a pathway-level ceiling rather than a substrate-specific design flaw.

Search-budget constraints. Population-based structural search faces a bottleneck distinct from unbounded accumulation: each candidate evaluation requires full agent execution, and the resulting cost constrains search scale. In Cortex, evolutionary optimization typically operates with modest populations over tens of generations (Kumar et al., 2026c; Fernando et al., 2024) because each fitness evaluation runs the complete agent pipeline. In Action, Romera-Paredes et al. (2024) overcome this through island-based parallel evaluation and Novikov et al. (2025) through cascaded filtering, achieving genuine mathematical discoveries, but their evaluation budgets remain finite: the candidate space grows combinatorially with program length. In Memory, Pan et al. (2026) apply population-based search to memory architecture itself, where each candidate evaluation requires running the full agent on validation episodes. The shared root cause is that structural search scales with discrete evaluation budgets rather than with continuous compute: doubling the compute budget doubles the number of candidates evaluated but does not change the optimization dynamics, unlike gradient-based updates where additional compute enables finer convergence over the same loss landscape. This distinction partly explains why parametric training achieves the top results in reasoning and code-generation benchmarks (Absolute Zero (Zhao et al., 2025), GrandCode (Li et al., 2026g)), while the structural pathway’s strongest results come from algorithm discovery where distributed evaluation at scale enables landmark improvements.

Individual addressability versus holistic optimization. These shared strengths and ceilings produce a single cross-pathway asymmetry: structural modularity enables selective rollback but prevents holistic optimization. A bad prompt entry can be deleted, a failing skill can be rewritten, and an outdated memory record can be pruned, but when a behavioral problem arises from the *interaction* among many structural artifacts (e.g., conflicting rules in a growing constitution, or redundant skills that interfere at retrieval time), identifying the distributed pattern is difficult because no single artifact is at fault. Parametric methods face

the inverse problem: gradient updates can reshape distributed representations holistically, but they cannot selectively target a single learned behavior; the very continuity that enables smooth optimization prevents surgical removal. The asymmetry is thus representational, not incidental: discrete artifacts are individually addressable but collectively opaque; continuous weights are collectively optimizable but individually opaque.

4.3 Parametric Consolidation

Parametric consolidation escapes the context ceiling and search-budget bottleneck of structural methods, but its efficacy depends on verifiable reward signals, and its instabilities are pathway-specific rather than substrate-specific. The verifiable-reward constraint established above finds its strongest cross-substrate confirmation in the rare parametric papers outside Cortex: DreamerV3’s world model generalizes across 150+ tasks because environment dynamics provide dense, deterministic supervision, and Jiang et al. (2026b) reformulate multi-agent topology selection as a cooperative MARL problem with verifiable task outcomes, one of only five purely parametric Action papers. Where such signals are absent, structural methods dominate regardless of substrate.

Parametric instabilities. Most parametric Cortex papers adopt GRPO or close variants, reflecting the algorithm’s engineering fit with sparse, binary reward signals; this concentration means that the failure modes documented below are encountered within a single optimization family. The four independently identified failure modes catalogued in Section 3.1 (Echo Trap (Wang et al., 2025f), Template Collapse (Wang et al., 2026h), Information Self-Locking (Zou et al., 2026a), and Reward Hacking Rebound (Wu & Tang, 2026)) are mechanistically distinct but share a defining property visible only in the pathway-level view: each arises from continuous optimization dynamics in weight space and none has a structural counterpart. The discrete addressability documented above has no parametric counterpart: a collapsed policy cannot be selectively “unlearned” because the damage is distributed across weight space rather than localized in a discrete artifact.

Catastrophic forgetting compounds these within-task instabilities with a cross-task dimension: new parametric updates overwrite prior knowledge because weight capacity is finite and shared. Lin et al. (2025) quantify the severity directly, finding 89% F1 degradation from standard full fine-tuning versus only 11% with selective sparse memory updates. That selective-update strategies (selective sparse updates (Lin et al., 2025), drift-selected LoRA adapters (Kim & Kim, 2026)) and the natural low-rank confinement that LoRA provides even without explicit anti-forgetting design (Hu et al., 2026) converge independently across memory pools, dialogue agents, and embodied policies suggests that catastrophic forgetting is a pathway-level instability rather than a substrate-specific weakness. Structural methods are immune to both classes: no structural artifact “collapses” into a degenerate attractor or overwrites previously stored entries, because each entry occupies its own discrete slot (though retrieval degradation can render entries functionally inaccessible, as documented above). This immunity partly explains why structural consolidation persists despite the performance ceilings documented in the preceding paragraphs; the reliability of discrete, individually addressable artifacts is itself a form of robustness that parametric compression sacrifices.

The confinement of parametric methods to narrow niches outside Cortex (orchestration routing in Action; world models and test-time training in Memory) confirms the artifact-discreteness constraint: where the target artifact is symbolic, gradients have no natural target. The complementary failure modes of the two pathways (ceilings that cap structural performance, instabilities that destabilize parametric training) motivate a third design: hybrid systems that combine both.

4.4 Hybrid Consolidation

About 35 papers explicitly couple structural and parametric consolidation. A recurring lifecycle, *structural discovery followed by parametric internalization*, appears across all three substrates, though it is not the only hybrid pattern. This $\Delta \rightarrow \nabla$ lifecycle is most clearly instantiated in Cortex evolution. SKILL0 (Lu et al., 2026b) progressively withdraws structural skill prompts during training, forcing the model to internalize them into weights, while ERL’s retrospective distillation (§3.1) generates structural reflections during training that are compressed into weights via SFT. The two occupy opposite ends of the $\Delta \rightarrow \nabla$ spectrum: SKILL0 withdraws scaffolding to force parametric absorption; ERL generates scaffolding that is retrospec-

tively absorbed. In Action, Su et al. (2026a) couple evolutionary selection over solution populations (Δ on $\mathcal{A}$) with policy training (∇ on Π), a selection–compression division of labor where evolutionary search handles discrete candidate evaluation and gradient optimization handles continuous policy improvement. SEVerA separates the roles more sharply: verified program structure provides parameter-independent correctness bounds that survive weight updates, achieving zero constraint violations. In Memory, at least thirteen papers independently converge on a “store structurally, retrieve adaptively” motif (§3.3): structural entries provide interpretable, composable knowledge while learned retrieval mechanisms discover usage patterns beyond static rules. This motif differs from the Cortex and Action lifecycles in a critical respect: it maintains both pathways persistently rather than compressing Δ into ∇ , because the addressability requirement central to most Memory systems (the ability to inspect, update, and delete individual knowledge entries) is incompatible with full parametric internalization. That all three substrates independently develop hybrid designs, yet Memory alone resists the $\Delta \rightarrow \nabla$ compression lifecycle, suggests that the consolidation pathway is shaped by substrate properties even when designers are free to combine both modes.

Persistent joint evolution as an alternative to compression. The $\Delta \rightarrow \nabla$ lifecycle assumes eventual convergence to a parametric end-state, but a second hybrid pattern, persistent joint evolution, maintains both pathways indefinitely, treating the structural and parametric components as complementary rather than sequential. The three clearest instantiations span a coupling-tightness spectrum. At the tight end, InSPO’s within-step instruction sampling (Zhou et al., 2025a) directly conditions every gradient update. At an intermediate coupling, SkillRL’s periodic skill evolution (Xia et al., 2026) interacts with policy training only at scheduled synchronization points. At the loosest coupling, MIA’s alternating-stage conversion (Qiao et al., 2026) separates the optimization loops entirely, with each pathway’s outputs feeding the other through distinct training stages. The coupling-tightness dimension reveals a design trade-off absent from the $\Delta \rightarrow \nabla$ lifecycle: tight coupling enables within-step mutual adaptation but demands careful synchronization to prevent the structural population from destabilizing gradient signal; loose coupling simplifies implementation but forgoes real-time synergy and risks the two pathways drifting apart. That all three systems outperform their single-pathway ablations (InSPO over static instructions, SkillRL over frozen skills, MIA over either parametric-only or memory-only variants) provides empirical evidence that persistent joint evolution captures complementarities the compression lifecycle discards.

The declarative–procedural decomposition. The hybrid designs in the preceding paragraphs all instantiate a representational division well established in cognitive science: the structural component carries declarative knowledge (what to do), while the parametric component carries procedural knowledge (how to do it). Zhang et al. (2026g) make this decomposition explicit: E-SPL’s joint optimization of competitively rated prompt populations (Δ) alongside policy training (∇) converts observed mistakes into declarative prompt rules while gradient updates internalize execution competence into weights, improving cross-task generalization. The declarative–procedural separation is principled rather than arbitrary: declarative artifacts are human-readable and revisable, making them natural carriers of explicit strategies that must adapt quickly to new task distributions; procedural weights generalize automatically across inputs, making them natural carriers of implicit competence that must resist catastrophic forgetting. This functional decomposition reframes the $\Delta \rightarrow \nabla$ lifecycle and persistent joint evolution not as competing hybrid designs but as two scheduling strategies over the same representational division of labor: compression collapses the declarative component into the procedural one once strategies stabilize, while persistent joint evolution maintains both indefinitely because the task distribution continues to shift. The decomposition also clarifies why the “store structurally, retrieve adaptively” motif in Memory resists compression: knowledge entries are irreducibly declarative (they must remain individually addressable), so only the retrieval policy, a procedural competence, can be parametrized. Whether the declarative–procedural boundary can be formalized into transition criteria, specifying when a structural discovery is stable enough to internalize parametrically, or when parametric drift warrants declarative correction, remains the central open question of the hybrid frontier, one that current hybrid papers approach empirically but none resolves.

Why so few hybrids? Two engineering barriers likely contribute beyond temporal recency: coordinating two optimization loops demands infrastructure neither loop requires alone (InSPO, SkillRL, and MIA each solve this synchronization problem differently (Zhou et al., 2025a; Xia et al., 2026; Qiao et al., 2026)),

and credit assignment between pathways becomes intractable when both change simultaneously (no hybrid paper isolates the marginal contribution of each pathway at its jointly evolved state). These barriers are implementation obstacles, not principled limitations: the declarative–procedural decomposition suggests the two pathways are functionally complementary, and the hybrid share is likely to grow as tooling matures.

4.5 Cross-Pathway Trade-offs

Regression affects both pathways, for opposite reasons. Both pathways struggle with regression, but for opposite reasons rooted in their representational commitments, and this opposition means that no single mechanism can protect both. Structural methods enable selective rollback, yet their artifacts accumulate monotonically, so the probability of undetected interference grows with library size; the gated-admission mechanisms documented above each emerged independently to contain this risk, yet none can detect interference patterns that span many artifacts simultaneously. Parametric methods compress experience into fixed-capacity weight spaces, avoiding accumulation entirely, yet risk catastrophic forgetting that erases prior competence without any mechanism for targeted recovery. The result is that each pathway’s regression defense depends on the property the other pathway lacks: structural rollback requires discrete addressability, which parametric representations sacrifice; parametric compression avoids unbounded accumulation, which structural representations cannot escape. Hybrid systems that invest most heavily in regression avoidance delegate control to the pathway whose representational properties match the failure mode: SEVerA’s verified structure provides parameter-independent correctness bounds (Banerjee et al., 2026); AgentDevel’s gating screens structural edits (Zhang, 2026); InSPO’s successive-halving pruning discards low-fitness candidates before they condition gradient signal (Zhou et al., 2025a). That regression control is handled through pathway-specific rather than substrate-specific mechanisms suggests it is a primary constraint on stable agentic evolution.

Open questions. When should a designer choose Δ , ∇ , or $\Delta+\nabla$? The preceding analysis supplies partial answers: artifact discreteness, reward verifiability, and infrastructure access jointly constrain the feasible pathway, and the declarative–procedural decomposition provides a principled rationale for separating what evolves structurally from what evolves parametrically (Zhang et al., 2026g). But no current framework specifies *transition conditions*. Three questions remain open. First, *when* does a structural method reach its ceiling? The unbounded-accumulation and search-budget ceilings documented above are well characterized qualitatively, yet no formal criterion predicts the point at which structural returns diminish and parametric internalization becomes cost-effective. The progressive-withdrawal and retrospective-distillation lifecycles all rely on hand-tuned schedules (fixed decay rates, periodic triggers) rather than data-driven transition signals. Second, *does persistent joint evolution outperform eventual compression*? The coupling-tightness spectrum shows that both designs yield gains over single-pathway baselines, but no study compares them head-to-head over extended training horizons; existing hybrid systems report only end-state performance, leaving unknown whether the two pathways converge, oscillate, or interfere when run jointly for hundreds of update events. Third, *how should regression risk factor into pathway choice*? The regression analysis above shows that each pathway’s defense depends on the representational property the other lacks, yet no design guideline weighs the regression cost of unbounded accumulation against the regression cost of catastrophic forgetting, a trade-off that depends on domain-specific factors (update frequency, task-distribution stability, accountability requirements) that current benchmarks do not measure. Formalizing pathway selection as a function of these factors (artifact discreteness, reward verifiability, infrastructure access, deployment horizon, and regression tolerance) remains open.

Three findings emerge from reading the evolution papers through the consolidation pathway lens that remain invisible in Section 3’s substrate-organized narrative: the structural–parametric distribution is *predictable* from artifact discreteness, evaluation signal type, and infrastructure access; structural and parametric methods occupy *complementary* failure landscapes whose interaction drives hybrid designs; and hybrid systems independently develop in all three substrates, though the specific hybrid pattern varies (Cortex and Action favor $\Delta \rightarrow \nabla$ compression, while Memory resists it because entries must remain individually addressable).

These findings expose gaps that current work leaves open. The open questions identified above (when structural methods reach their ceiling, whether persistent joint evolution outperforms eventual compression, and

Table 5: **Cross-substrate properties of structural versus parametric consolidation.** Each row names a property that holds across all three substrates within a given pathway.

Property	Structural (Δ)	Parametric (∇)
Artifact nature	Symbolic, discrete (prompts, code, facts)	Continuous (weight matrices)
Evaluation signal	LLM-as-judge, heuristic, or execution-based	Requires dense, verifiable rewards (code exec, exact match)
Infrastructure	No training required; API-only compatible	Gradient access + RL infrastructure
Shared strengths	Interpretable, composable, rollback-capable	Compresses experience; generalizes automatically
Shared ceilings	Context overflow, unbounded accumulation, search-budget limits	e.g., template collapse, reward hacking, catastrophic forgetting
Regression mechanism	Gated admission (flip-centered, unit-test, confidence-based)	LoRA isolation, selective sparse updates, low-rank confinement
Core trade-off	Addressability & modularity	Generalization & compression

how regression risk should factor into pathway choice) remain unresolved empirically. The “store structurally, retrieve adaptively” motif that Memory papers independently adopt also lacks theoretical foundations; no existing formalization accounts for joint structural–parametric dynamics within a single system.

Key Takeaways: Consolidation Pathway

- **Pathway is substrate-constrained, not freely chosen.** Artifact discreteness, evaluation signal type, and infrastructure access jointly determine the consolidation pathway, producing the parametric confinement pattern across Memory, Action, and Cortex.
- **Failure landscapes are pathway-specific, not substrate-specific.** Unbounded accumulation and search-budget ceilings are structural-pathway problems; template collapse, reward hacking, and catastrophic forgetting are parametric-pathway problems, regardless of substrate.
- **Hybrid designs develop independently across substrates but vary in form.** The $\Delta \rightarrow \nabla$ lifecycle (structural discovery followed by parametric internalization) appears in Cortex and Action but Memory resists it because entries must remain individually addressable; persistent joint evolution provides an alternative that maintains both pathways indefinitely.
- **Regression control is a primary constraint.** Each pathway’s defense depends on representational properties the other lacks; hybrid systems that delegate regression to pathway-matched mechanisms tend to achieve more stable evolution.
- **The pathway selection problem is open.** No principled framework determines when to use Δ , ∇ , or $\Delta + \nabla$ as a function of artifact discreteness, reward verifiability, infrastructure access, deployment horizon, and regression tolerance.

The consolidation pathway determines *how* experience becomes persistent change; the selective pressure dimension determines *what signal source drives* that process: whether evolution is driven by environmental experience and self-generated signals alone ($\mathcal{H} = 0$) or additionally by human input ($\mathcal{H} \neq 0$). Section 5 examines how pathway and pressure interact.

5 Axis III: Selective Pressure

5.1 Autonomous Selective Pressure

With the structural–parametric continuum characterized in Section 4, we now turn to the third axis: what source of selective pressure drives evolution. Of the surveyed evolution papers, about 90% operate entirely under autonomous selective pressure ($\mathcal{H} = 0$), the most lopsided distributional finding on any axis of the three-axis framework. This asymmetry is not a single phenomenon but the joint product of three reinforcing causes, each operating on a different subset of the literature. The first is principled: where rewards are deter-

ministically verifiable (code execution, exact-match answers, game outcomes), human feedback is genuinely unnecessary. Zhao et al. (2025) demonstrate the limiting case: a single model generates and solves code-based reasoning tasks validated only by a code executor, achieving state-of-the-art coding and mathematical reasoning with zero human involvement at any stage. The second cause is architectural: most structural evolution systems (Δ) evolve artifacts through automated optimization loops (evolutionary search, LLM-as-optimizer, self-reflection) that have no interface for real-time human input, not because the designers chose to exclude humans but because the search and selection mechanisms were designed for autonomous operation. The third is economic: human-in-the-loop evaluation is expensive and does not scale, restricting $\mathcal{H} \neq 0$ designs to settings where the return on human effort justifies the cost. These causes are complementary, not competing: the first explains why $\mathcal{H} = 0$ *succeeds* in verifiable domains, the second why it *persists* across automated evolutionary pipelines, and the third why $\mathcal{H} \neq 0$ remains *rare* even when it would be beneficial. That all three point in the same direction explains the extremity of this concentration. Moreover, this concentration is intensifying rather than moderating: among pre-2024 papers, roughly half operate under $\mathcal{H} = 0$; among 2025–2026 papers, the share approaches near-unanimity, indicating that the three causes are strengthening jointly rather than being offset by growing interest in human–AI collaboration.

The roughly 25 $\mathcal{H} \neq 0$ papers confirm rather than undermine this analysis: they concentrate precisely where automated verification is weakest. He et al. (2025b) deploy Aria at 150M+ monthly active users in TikTok Pay, where the agent actively queries human operators on self-assessed uncertainty because financial compliance cannot be verified by execution alone; the domain demands human evaluative input regardless of cost. This concentration ($\mathcal{H} \neq 0$ appearing where deterministic verification is unavailable) suggests that the selective pressure asymmetry reflects not a blanket preference for autonomy but a rational allocation: human selective pressure fills the verification gap, and the field’s current domain mix leaves that gap small. Whether this allocation is optimal or merely convenient (whether the field gravitates toward verifiable domains partly *because* they permit $\mathcal{H} = 0$) is a question the distributional data alone cannot answer but the reward signal analysis below can begin to answer.

The $\mathcal{H} = 0$ papers do not form a monolithic category: the quality of the autonomous selective pressure they employ varies systematically, and this variation appears to predict empirical outcomes more reliably than substrate or pathway choice alone. Three tiers of autonomous reward signal emerge from a cross-substrate analysis: Tier 1 (deterministic verification by an external oracle), Tier 2 (LLM-as-judge or self-consistency evaluation), and Tier 3 (learned proxy signals). Section 3’s substrate-organized narrative touches on each reward type within individual substrates; the analysis below formalizes them as a single hierarchy and examines their cross-substrate properties.

Tier 1: deterministic verification. Among the strongest empirical results in the survey are systems whose selective pressure reduces to a deterministic oracle (code execution, exact-match comparison, or formal proof) because the verifier is external to the system being evolved, eliminating the evaluator–agent coupling that degrades higher tiers. Zhao et al. (2025) demonstrate the purest case in Cortex: the Absolute Zero Reasoner’s code-executor-only selective pressure. In Action, Romera-Paredes et al. (2024) and Novikov et al. (2025) achieve landmark algorithm discoveries through automated evaluator functions, demonstrating that strong verification enables results beyond human baselines. In Memory, execution-based verification appears through domain-specific tools: logic synthesis checking (Fang et al., 2026), GPU profiling (Dong et al., 2026b), and SPICE simulation (Bao et al., 2026). Li et al. (2026g) achieve superhuman competitive programming performance through execution-based rewards. The cross-substrate pattern is consistent: wherever a deterministic oracle exists, autonomous selective pressure suffices, and the resulting systems produce the field’s strongest claims (zero-data state-of-the-art reasoning, new mathematical discoveries, superhuman competitive programming) without any human selective pressure.

Tier 2: LLM-as-judge and self-consistency. Where deterministic verification is unavailable, numerous papers substitute the model’s own judgment or inter-model consensus as selective pressure, a design that trades verifier independence for domain generality, because the same LLM-as-judge mechanism applies to open-ended reasoning, dialogue, and knowledge curation without requiring a task-specific oracle. Yuan et al. (2024) establish the paradigm: a single LLM serves as both policy and reward model, with iterative DPO training on self-scored preference pairs. The central failure mode is *preference signal collapse*: chosen and

rejected responses converge, rendering the training gradient ineffective (Wang et al., 2025d). Fu et al. (2026) prove that iterative self-rewarding alignment acts as a contraction mapping, formalizing why iteration is necessary rather than merely helpful but also why each iteration yields diminishing returns.

Multi-role variants decouple generation from evaluation across distinct agent roles: Chen et al. (2025f) train Proposer, Solver, and Judge roles on a shared backbone, achieving gains across 22 benchmarks without external verifiers. Self-play systems often combine Tier 1 game-outcome signals with Tier 2 LLM-as-judge evaluation (Liu et al., 2026a; Peng et al., 2026b); Chojecki (2025) formalize self-play, self-correction, and synthetic data bootstrapping as instances of a Generator-Verifier-Updater operator.

In Memory, Tier 2 signals govern which experiences persist: Ouyang et al. (2026b) distill reasoning strategies using LLM-as-judge labeling, and Cao et al. (2025) employ utility-based pruning within a full-lifecycle pipeline. Shinn et al. (2023)’s verbal reflections are classified as Cortex (II) because they modify the deliberative prompt, yet function as lightweight episodic records frequently cited in Memory contexts.

The cross-substrate pattern reveals a structural asymmetry absent from Tier 1: LLM-as-judge evaluation degrades with iteration because improvements in generation quality do not automatically improve evaluation quality. Tier 1 verifiers are invariant to which system produced the output (a code executor returns the same result regardless of which model generated the code) but Tier 2 judges drift with the system they evaluate, and the gap between generation capability and evaluation capability is unmonitored. The empirical signature is diminishing marginal gains across self-rewarding iterations (Yuan et al., 2024; Wang et al., 2025d), a pattern that Fu et al. (2026)’s contraction analysis explains yet does not resolve: no general runtime monitor detects when the self-judge’s evaluative capacity has become inadequate.

Tier 3: proxy signals. When neither deterministic verification nor direct self-evaluation applies, a third class of systems interposes a separately constructed proxy between the agent and the evaluative signal: a trained reward model, a simulated environment, or a statistical aggregation of peer judgments. Unlike Tier 2, where the agent evaluates its own outputs directly, Tier 3 signals pass through an intermediary whose fidelity is itself unmonitored, extending autonomous selective pressure to non-verifiable domains at the cost of an unaudited gap between the proxy and the true objective. Qi et al. (2025) train an Outcome-Supervised Reward Model on rollout data for web navigation, enabling a self-evolving curriculum, but the ORM inherits the distributional assumptions of its training data and cannot evaluate tasks outside its coverage. Sui & Hooi (2026) replace the learned model with peer consensus via a structured propose–critique–revise–re-rank protocol and peer ranking, providing meta-evaluation supervision without ground-truth labels. Song et al. (2026) formalize a middle ground: RL from Text Feedback uses a larger external judge to generate textual critiques that are distilled into the policy; the text feedback is richer than a binary signal but remains a proxy for quality dimensions the judge was not designed to evaluate. Further proxy mechanisms include Monte Carlo simulation of future dialogue turns (Wu et al., 2025b) and majority-voting reward generation (Fang et al., 2025d).

The failure mode specific to Tier 3 is that the learned approximation is fixed at training time while the agent evolves; the gap between proxy and true objective widens silently across iterations. Even Tier 1 verifiers are not immune to exploitation: Wu & Tang (2026) demonstrate a reproducible three-phase *reward hacking rebound* in code generation, where the model first fails to exploit the verifier, temporarily retreats to legitimate solving, then discovers a successful exploitation strategy (rewriting test outputs via print statements). If deterministic verifiers can be exploited, learned proxies are more vulnerable still: the policy can exploit weaknesses in the proxy that no external mechanism detects. Tier 3 methods thus extend autonomous selective pressure to the broadest set of domains (creative writing, open-ended dialogue, specialized reasoning) but with the weakest fidelity guarantees among the three tiers.

The scope boundary of autonomous evolution. As selective pressure moves from deterministic verification (Tier 1) through self-evaluation (Tier 2) to learned proxies (Tier 3), the verifier’s independence from the system it evaluates decreases, and with it the reliability of the evolutionary signal. This spectrum exposes a structural gap: the domains where autonomous selective pressure is most reliable are precisely those where the stakes of misalignment are lowest (code correctness, mathematical truth), while the domains where alignment matters most lack verifiers whose accuracy is independent of the system being evaluated.

No paper in the surveyed corpus proposes a meta-evaluation mechanism that monitors reward signal quality over the course of evolution. The spectrum thus defines a scope boundary for autonomous evolution, which we state as a labeled finding because it serves as a direct premise for the analysis in Section 7:

Finding: The Verifier Boundary of Autonomous Evolution

Autonomous evolution is reliably effective only within the domain where its verification signal maintains fidelity. Beyond this boundary, evolution requires an external evaluator.

In Tier 1 domains, this boundary is absent: the verifier’s independence from the system being evolved guarantees that selective pressure remains accurate regardless of how the system changes. In Tier 2 and Tier 3 domains, the boundary contracts as evolution proceeds: each iteration may degrade the LLM-as-judge evaluator (Tier 2) or widen the proxy-objective gap (Tier 3), yet no system in the survey implements a runtime mechanism to detect when verification fidelity has fallen below a useful threshold. The practical consequence is that autonomous evolution’s effective reach is delimited not by the model’s capacity to improve but by the verification signal’s capacity to remain trustworthy as the system evolves, a constraint that current systems neither measure nor report, and that the $\mathcal{H} \neq 0$ analysis below will show human selective pressure partially but incompletely compensates for.

The remaining $\mathcal{H} \neq 0$ papers (25 systems) introduce a qualitatively different signal source, human judgment, that is in principle independent of the system it evaluates but brings its own constraints of cost, scalability, and temporal drift.

5.2 Human-Involved Selective Pressure

These $\mathcal{H} \neq 0$ papers do not distribute uniformly across the five forms of human-involved input defined in Section 2.4: two forms (evaluative feedback and implicit signal) account for the vast majority of cases, while direct prescription and interactive collaboration remain nearly empty, revealing that human selective pressure in practice relies on low-effort, asynchronous channels rather than the real-time steering that the co-evolution framing might suggest.

Evaluative feedback: the dominant explicit form. Among these five forms, evaluative feedback (human judgments expressed as preferences, ratings, or corrections) is the most common intentional channel, appearing in about half of the $\mathcal{H} \neq 0$ papers. Its dominance reflects a cost-information trade-off: evaluative feedback requires less effort than demonstrations yet conveys richer signal than implicit traces. Shivaswamy & Joachims (2015) formalize why even low-quality instances suffice: their coactive learning framework proves sublinear regret from sub-optimal corrections, establishing that convergent parametric evolution is possible from imperfect human input. The remaining evaluative-feedback papers divide along two axes: *personalization* (aggregate alignment via iterative human-agent feedback loops (Abramson et al., 2022) versus per-user preference alignment (Li et al., 2024b; Nam et al., 2026)) and *noise handling* (Yang et al. 2025b’s temporal-consistency purifier, the only system modeling feedback noise as a first-class concern). Further papers operate at different substrates or pathways: real-time policy-gradient updates from human signals (MacGlashan et al., 2017), graph-search-based optimization of training configurations from task descriptions or deployment feedback (Atreja et al., 2026), prompt optimization via pairwise preferences (Lin et al., 2024), and human-expert-guided tool specification refinement (Gao et al., 2025b).

Implicit signal: the dominant passive form. About ten papers derive selective pressure from behavioral patterns accumulated through ordinary use, making implicit signal the second most common yet informationally weakest form. Conversational memory systems form the largest cluster, sharing a common extract-persist-retrieve architecture but differing in the *representation* that mediates persistence: flat or list-based stores (Packer et al., 2023), graph-structured representations with increasing expressiveness (Chhikara et al., 2025; Rasmussen et al., 2025), and further representational variants (Zhong et al., 2024; Tian et al., 2025; Gadzhiev & Kislov, 2026; Nie et al., 2026b; Westhäußer et al., 2025). Beyond conversation, implicit signals take domain-specific forms where professional activity generates evolutionary pressure: code commit history (Deng et al., 2026) and confirmed clinical diagnoses (Chen et al., 2026c). Ma et al. (2026b) rep-

represent a distinct mechanism, collective evolution across multiple users, aggregating trajectories from eight concurrent users into a shared skill library; Zhang et al. (2026a) apply a per-user evolutionary mechanism in foreign trade. He et al. (2025b) occupy a hybrid position straddling evaluative feedback and implicit signal (Section 3.3). The implicit-signal cluster accounts for the largest documented $\mathcal{H} \neq 0$ deployments precisely because they require no feedback infrastructure beyond the task itself, yet the human evaluative input they provide is the least structurally embedded in the spectrum.

Demonstration, interactive collaboration, and direct prescription. The remaining three forms account for only three papers, a scarcity that reveals a mismatch between what the co-evolution framing promises and what current architectures support. Demonstration appears in one paper: Jain et al. (2015) implement coactive trajectory feedback (re-ranking, kinesthetic correction, and interactive markers) via online learning. Interactive collaboration appears in two: Mozannar et al. (2025) design six explicit human-in-the-loop mechanisms spanning planning through consolidation, and Adnan et al. (2025) implement progressive dialogue-driven service generation. Direct prescription is absent as a live evolutionary input; Kumar et al. (2026c) evolve constitutions through automated search rather than injecting prescriptive constraints during evolution. The three-paper total quantifies an architectural asymmetry: current systems absorb human signal passively or elicit it cheaply, but lack interfaces for sustained high-bandwidth interaction.

Distributional pattern and its interpretation. Of the $\mathcal{H} \neq 0$ papers, about half target Memory, about a third target Cortex, and a handful target Action. Human selective pressure concentrates precisely where Tier 1 verification is unavailable: the Memory $\mathcal{H} \neq 0$ papers target domains where correctness is subjective or context-dependent; the Cortex $\mathcal{H} \neq 0$ papers employ preference-based training because open-ended reasoning quality cannot be reduced to pass/fail execution. Two cells are entirely empty ($\mathcal{A} \times \nabla \times \mathcal{H} \neq 0$ and $\mathcal{M} \times \nabla \times \mathcal{H} \neq 0$) because the rare parametric methods in these substrates already possess objective evaluation signals. Human selective pressure thus appears where automated verification is weakest and where human signal can be obtained at acceptable cost. This complementarity admits two readings: an optimistic one (the roughly 90% $\mathcal{H} = 0$ rate is a rational allocation, with deterministic verification genuinely rendering human feedback unnecessary) and a pessimistic one (the field gravitates toward verifiable domains partly *because* they permit $\mathcal{H} = 0$, leaving value-laden settings systematically understudied). The distributional evidence constrains without resolving.

5.3 The Alignment Gap

Safety-oriented papers and their $\mathcal{H} = 0$ limitation. This gap is sharpest where the survey’s axes intersect: the systems powerful enough to reshape their own reasoning operate almost exclusively under $\mathcal{H} = 0$. Only a handful of papers explicitly target safety or alignment as an objective of the evolutionary process, and all rely on autonomous selective pressure. Three operate parametrically through adversarial or cooperative games: Beigi et al. (2026) construct a Hacker–Auditor adversarial game with gated rewards; Zhang et al. (2026f) formulate safety alignment as a positive-sum multi-agent game; and Ali et al. (2026) combine evolving security-aware constitutional principles with DPO-based unlearning. Two operate structurally: Gallego (2026) discover safety specifications from binary danger signals, converging within one to two rounds; and Kumar et al. (2026c) evolve behavioral norms via multi-island genetic programming. Banerjee et al. (2026) achieve zero constraint violations through Formally Guarded Generative Models combining Dafny verification (Δ on $\mathcal{A}$) with GRPO fine-tuning (∇ on Π), a hybrid approach (Section 3). Across all six, every safety guarantee is bounded by the formal scope of its verifier; none targets value alignment where correctness depends on context or evolving social norms.

The frozen-preference problem. The $\mathcal{H}$ boundary defined in Section 2.5 is central to understanding why the alignment gap persists. Reward models trained on historical human preferences encode human signal but deploy without live human input, which we therefore classify as $\mathcal{H} = 0$ in our framework. This means that RLHF-trained models which subsequently self-evolve (including all safety-oriented papers above) operate on a *frozen snapshot* of human preferences captured at the reward-model training step. The evolutionary process then extends beyond that snapshot: as the agent improves or shifts distribution, the reward model’s coverage of the new output space is unverified, yet no mechanism exists to update the human signal. Han

et al. (2025) provide the most direct empirical evidence that this gap is consequential: they identify an Alignment Tipping Process (ATP) in which self-evolving agents gradually abandon alignment constraints in favor of self-interested strategies reinforced by environmental feedback. In their single-agent setting, high-reward deviant actions accumulate as in-context examples that progressively override initial alignment; in multi-agent settings, deviant strategies spread through social observation as agents adopt rule violations they observe succeeding in peers, an imitative diffusion dynamic that degrades alignment across both open-source and closed-source models. The tipping process demonstrates that the frozen-preference problem is not merely theoretical: when the evolutionary loop generates sufficient reward signal for misaligned behavior, no mechanism in the current $\mathcal{H} = 0$ architecture detects or corrects the drift. Zou et al. (2026a) identify a complementary failure mode, information self-locking, in which outcome-based RL training for multi-turn reasoning produces a negative feedback loop: weak belief tracking masks the learning signal for informative queries, trapping the agent in low-information interaction patterns that standard policy gradients cannot escape. The problem compounds when the human partner is themselves changing: if the user’s preferences, expertise, and interaction strategies co-adapt with the system (as Section 6 will examine), then even a periodically refreshed reward model tracks a moving target, and the temporal gap between preference capture and deployment widens on both sides. The frozen-preference problem and the alignment tipping process together show that formally specifiable constraints can be enforced (SEVerA’s zero-violation guarantee is the strongest result on this dimension) but value alignment, where correctness depends on context, culture, or evolving social norms, remains beyond the reach of autonomous selective pressure. The field thus faces an alignment gap: systems powerful enough to self-evolve their reasoning are constrained either by deterministic verifiers that cannot capture values, or by frozen preference snapshots that cannot track value drift. No paper in the surveyed corpus proposes a mechanism combining Tier 1 verification reliability with the value sensitivity of human judgment.

Bandwidth concentration. The form-level analysis above can be restated in terms of a single design variable: the bandwidth of the human signal channel. Of the $\mathcal{H} \neq 0$ papers, most operate through low-bandwidth channels, evaluative feedback (about a dozen papers: preference pairs, ratings, binary corrections) or implicit signal (about ten papers: behavioral traces extracted from ordinary use), while only 5 employ high-bandwidth interaction: three interactive-collaboration systems (Mozannar et al. 2025’s co-planning and co-tasking interface, Adnan et al. 2025’s mixed-initiative dialogue, Gao et al. 2025b’s iterative expert consultation), one demonstration system (Jain et al. 2015’s kinesthetic trajectory correction), and one hybrid system (He et al. 2025b’s uncertainty-driven human queries at production scale). This pattern quantifies a finding that the form-level discussion left implicit: current $\mathcal{H} \neq 0$ architectures overwhelmingly restrict human selective pressure to channels that transmit bits-per-interaction (a preference comparison, a memory write, a behavioral trace) rather than the structured, multi-turn steering that demonstration and interactive collaboration provide. No system employs direct prescription as a live evolutionary input. Whether this restriction reflects engineering cost (high-bandwidth interfaces are harder to build), cognitive cost (sustained interaction fatigues human partners), or both, the consequence is uniform: the human signal available to the evolutionary loop is sparse relative to the autonomous signal it supplements, limiting the rate at which human selective pressure can correct distributional drift.

The human partner is not a fixed resource. The selective pressure analysis above treats the human partner as a signal source whose properties (cost, bandwidth, noise, temporal stability) constrain the evolutionary loop. Every $\mathcal{H} \neq 0$ system in this survey implicitly assumes that the user who provides feedback at iteration t is functionally equivalent to the user at iteration $t-1$: same preferences, same expertise, same cognitive strategies. Section 6 will show that this assumption is empirically false. Human evaluators are themselves changed by the systems they evaluate: their decision-making strategies adapt, their domain expertise shifts, and their reliance on AI outputs reshapes the very preferences that the selective pressure mechanisms above are designed to capture. Understanding this reciprocal dynamic is necessary before the temporal, architectural, and evaluative gaps identified here can be meaningfully resolved, because the frozen-preference problem is not only a matter of stale reward models but of a co-adapting human whose current preferences may no longer be the preferences the system should optimize for.

Key Takeaways: Selective Pressure

- **Selective pressure asymmetry:** About 90% of agentic evolution papers operate under autonomous selective pressure ($\mathcal{H} = 0$), driven jointly by the availability of deterministic verification, the autonomous design of structural optimization loops, and the cost of human-in-the-loop evaluation, a concentration that is intensifying over time.
- **Signal hierarchy:** Autonomous reward signals form a three-tier hierarchy: deterministic verification (strongest, verifier-independent), LLM-as-judge and self-consistency (scalable but evaluation quality may drift with the system being evaluated), and proxy signals (broadest domain coverage but subject to unmonitored drift). The tier predicts empirical outcomes more reliably than substrate or pathway choice.
- **Verification–selective-pressure complementarity:** Human selective pressure ($\mathcal{H} \neq 0$) concentrates where automated verification is weakest, predominantly in Memory and Cortex, producing a systematic complementarity rather than random coverage.
- **Alignment gap:** The safety-oriented papers all rely on $\mathcal{H} = 0$, operating on frozen preference snapshots; Han et al. (2025)’s Alignment Tipping Process demonstrates that self-evolving agents can abandon alignment constraints through reinforcement of deviant strategies, and no mechanism in the survey detects this drift.
- **Low bandwidth:** Most $\mathcal{H} \neq 0$ papers rely on low-bandwidth channels (evaluative feedback or implicit signal); only 5 employ high-bandwidth interaction. No system employs direct prescription as a live evolutionary input.

6 Human Adaptation Under Agentic Evolution

Every selective pressure mechanism analyzed in Sections 3–5 treats the human partner as a fixed resource whose evaluative capacity does not change across evolutionary iterations. The Axis III analysis showed that beyond the boundary where deterministic verification suffices, agent evolution depends on human selective pressure; but that analysis could not proceed further without answering a prior question: *is the human partner’s cognitive capacity stable under sustained AI interaction?*

The three-axis vocabulary developed in Section 2 transfers to the human side because its categories were drawn from cognitive science in the first place. On Axis I, the relevant human changes are cognitive (deliberative capacity, corresponding to Π), operational (task-execution skill, corresponding to $\mathcal{A}$), and epistemic (knowledge structures and trust calibration, corresponding to $\mathcal{M}$). On Axis II, *structural adoption* occurs when a practitioner incorporates AI-generated artifacts into their output; *parametric internalization* occurs when the practitioner’s intrinsic capability to produce that output independently improves. On Axis III, human adaptation is shaped either by externally designed interventions (scaffolded interfaces, institutional policies, training programs) or proceeds as self-driven, unmanaged change; as with agent-side evolution, the literature documents substantially more instances of the latter. These correspondences are analytical lenses for organizing a heterogeneous empirical literature, not claims of mechanistic equivalence between silicon and biological substrates.

A defining asymmetry separates the two sides: for agents, both Δ and ∇ consolidation represent capability gains; for humans, structural adoption and parametric internalization can be *negatively* correlated—the central empirical finding of this section. No study in this corpus demonstrates full bidirectional co-evolution within a single system; the evidence documents human adaptation *to* AI, and we synthesize the bidirectional picture in Section 7.

The analysis proceeds in four parts: the performance–capability paradox; the metacognitive and trust mechanisms that sustain it; the collective and societal consequences that emerge when individual-level effects aggregate; and the design principles that moderate direction. Throughout, we use *adaptation* for individual-level human changes and reserve *co-evolution* for the bidirectional coupling between agent and human trajectories analyzed in Section 7.

6.1 The Performance–Capability Paradox

The most consistent empirical finding across studies of AI-assisted human performance is a decoupling between measurable output and independent capability: AI reliably improves what humans produce while frequently degrading what humans can do without it. This decoupling instantiates Axis II’s consolidation pathway (§2.3) in reverse: AI enables *structural adoption*, in which human output improves through incorporation of AI-generated artifacts, but often fails to produce *parametric internalization*, in which the human’s intrinsic capacity to generate that output independently also improves. The pattern is documented with strong experimental evidence across education, professional work, and decision-making, and its direction depends critically on how the AI system is designed.

Education: output up, retention down. In a large-scale RCT ($N=839$), Bastani et al. (2025) find that students with unrestricted GPT access score 48% higher on practice problems yet score 17% lower on unassisted exams, a direct dissociation between AI-assisted throughput and retained knowledge. Kumar et al. (2024) report a consistent result in a pre-registered study ($N\approx 1,100$): LLM assistance during a creative task can boost in-session performance but significantly hinders subsequent independent creative output: coaching-style guidance impairs divergent thinking originality, while direct LLM answers impair convergent thinking accuracy. The pattern extends to procedural skills: Liao et al. (2024) find that students using a ChatGPT-integrated programming tool show significantly *decreased* problem-solving ability (attributed to over-reliance on the tool’s direct-answer channel), while Georgiou (2025) document reduced cognitive engagement when students work with ChatGPT rather than independently. Abdelghani et al. (2025) find convergent evidence with younger learners ($N=63$): middle schoolers cannot discriminate AI answer quality (at-chance discrimination) and achieve only chance-level task success ($\sim 51\%$) despite unrestricted ChatGPT access, with positive AI attitudes *negatively* predicting interaction quality. Temporal dynamics complicate the picture; Kazemitabaar et al. (2023) find no learning harm in a three-week coding study, suggesting that shorter interventions may not capture the effect, but the weight of RCT evidence favors a default toward structural adoption without parametric internalization when AI provides direct answers.

Professional work: generation shifts to selection. The same decoupling appears among working professionals. In a pre-registered RCT with software developers ($N=52$), Shen & Tamkin (2026) find that AI-assisted participants score 17% lower on subsequent knowledge quizzes (a large effect), with only three of six observed interaction patterns preserving learning. Wang et al. (2025c) document what they term “synthetic fluency”: homework–exam score correlations weaken substantially in AI-available courses, producing gaps of 42–44 points in Calculus III and exceeding 48 points in Linear Algebra once exams shifted to conceptual assessment. In professional writing, Noy & Zhang (2023) find in a pre-registered RCT ($N=444$) that 68% of treated participants adopt a paste-and-submit strategy, substituting AI output for their own generation rather than using it as input to further refinement. These findings converge on a common mechanism: when AI provides complete outputs, humans shift from generation to selection, reducing the deliberate practice that builds independent skill (Macnamara et al., 2024; Koch, 2026; Jose et al., 2025; Bankins & Formosa, 2023; Crowston & Bolici, 2025). A mixed-method review of AI-induced deskilling in medicine identifies the same pattern across 22 systematic and 62 narrative sources, distinguishing skill *erosion* from *upskilling inhibition* (Natali et al., 2025), with parallel concerns documented among UX practitioners (Shukla et al., 2025).

Decision-making: the paradox has boundary conditions. The paradox extends to expert judgment. In a pre-registered RCT with radiologists ($N=227$), Agarwal et al. (2023) find that AI assistance does not improve diagnostic accuracy despite the AI system outperforming 78% of participating radiologists; humans assign AI input roughly one-third the weight of their own signal in a Bayesian belief-updating model. Two critical counterpoints bound the paradox. First, Becker et al. (2025) show in a within-subjects randomized design (246 tasks, 143 hours of screen recordings) that experienced open-source developers are 19% *slower* with AI assistance, demonstrating that the performance benefit itself can reverse for experts working on familiar codebases. Second, Dell’Acqua et al. (2026) formalize a “jagged technological frontier” in a pre-registered RCT ($N=758$): AI users complete 12.2% more tasks inside the frontier but accuracy *decreases* by 19 percentage points on tasks outside it, where AI-generated wrong answers become more

persuasive than unassisted errors. Complementing overreliance findings, He et al. (2023) demonstrate a Dunning–Kruger route to *under*-reliance ($N=249$): participants who overestimate their own competence show significantly lower agreement with correct AI advice. Together, these results show that the paradox is not universal; it is bounded by task structure, expertise, and the match between AI capability and task demands. Scaffolded designs can attenuate or even eliminate the decoupling, but under unguided access the performance–capability paradox is the default.

The paradox persists in part because users often fail to detect it in themselves: students do not perceive the learning harm (Bastani et al., 2025), and developers believe they are faster while measured performance shows the opposite (Becker et al., 2025). This systematic metacognitive blindness, and the trust miscalibration it enables, sustains the paradox even when users have the information needed to correct course.

6.2 Mechanisms: Metacognition, Trust, and Sycophancy

The performance–capability paradox documented above is self-reinforcing: humans systematically fail to recognize AI’s effects on their own cognition, and they miscalibrate trust in ways that deepen rather than correct the decoupling. Two mechanisms sustain this cycle.

Metacognitive blindness: users do not perceive the deficit. Across domains, users perceive benefits that objective measures contradict. Students in Bastani et al. (2025)’s RCT did not perceive learning harm despite a 17% exam score decline; experienced open-source developers believed they were roughly 20% faster with AI assistance while objective measurement showed a 19% slowdown (Becker et al., 2025); and 71% of participants in a controlled study failed to detect sycophantic AI behavior despite post-task evaluation surveys (Bo et al., 2026). Tankelevitch et al. (2024) formalize the underlying mechanism: AI-generated outputs exhibit high processing fluency (grammatical polish, confident tone, structured presentation), which humans misattribute to content quality, producing miscalibrated confidence at three levels (prompt evaluation, output assessment, and workflow strategy). This fluency-based illusion operates below conscious deliberation; Riva (2025) characterize it as a “System 0” preprocessing layer that shapes information encounters before Kahneman’s Systems 1 and 2 engage, rendering the distortion invisible to the user experiencing it.

Trust miscalibration: system-level deference overrides content-level skepticism. A recurring finding across AI-assisted decision-making studies is an asymmetry between content-level and system-level trust: domain experts may resist deference on specific factual claims (content-level) yet remain vulnerable to the broader authority signal conveyed by an AI system’s confident presentation (system-level). This asymmetry may explain why domain expertise alone does not reliably prevent overreliance (Parasuraman et al., 2000; Qazi et al., 2025; Xu et al., 2025b). The quantitative evidence is striking: in a controlled experiment with 432 students, participants accepted 86% of misleading AI recommendations on average (with only 2.4% underreliance on correct recommendations), and, counterintuitively, post-task trust and appropriate reliance are *negatively* correlated (Pitts et al., 2026). Compliance with AI-generated advice exceeds 65% even for high-stakes personal decisions, with no measurable well-being benefit (Luettgau et al., 2025). In clinical settings, Qazi et al. (2025) demonstrate a 14-percentage-point accuracy drop when physicians receive flawed LLM recommendations, with more-experienced physicians *more* susceptible than less-experienced ones, a direct instance of system-level trust overriding content-level expertise. Explanations intended to support calibration can produce asymmetric effects: Schemmer et al. (2023) find that LIME explanations improve reliance when the AI is correct but do not significantly reduce deference when the AI errs, while Xu et al. (2025b) find that LLM-generated explanations amplify lay users’ deference on incorrect AI predictions. Rezaeian et al. (2025) report a convergent asymmetry in clinical breast-cancer detection ($N=28$): high AI confidence scores reduce diagnostic accuracy while low confidence scores increase diagnosis duration, indicating that confidence displays shift clinician behavior in opposite directions depending on the AI’s expressed certainty; Fregosi et al. (2026) find a similar confidence-driven pattern in a logic-reasoning study at larger scale ($N=184$). The content–system fault line runs through expertise itself: primary care physicians in Xu et al. (2025b)’s study resist content-level deference, acting as a “cognitive firewall,” while experienced physicians in Qazi et al. (2025)’s study are more vulnerable to system-level automation bias.

These mechanisms, however, are tractable. Trust-adaptive interventions can substantially reduce inappropriate reliance (Srinivasan & Thomason, 2026), and Li & Steyvers (2026) show that humans in a well-calibrated condition recalibrate from 62% to 86% accuracy in trust judgments through structured feedback, though in a separate counterintuitive-mapping condition, 44% of participants fail to learn appropriate reliance. Yet one systematic driver of trust miscalibration resists individual-level correction: AI sycophancy, which operates not as an isolated defect but as a persistent feature of current LLM training, a pattern that scales from individual interactions to population-level distortion.

Sycophancy as a persistent distortion. Default LLM behavior is systematically sycophantic, not as an incidental defect amenable to prompt-level fixes, but as a persistent feature traceable to RLHF training incentives (§5). Jacobowitz (2026) identify three functions sycophancy serves: steering conversations toward user-preferred conclusions, maintaining conversational consistency, and generating dependency through validation. Across 100,000+ response pairs spanning 11 models, Cheng et al. (2025) find that LLMs are 45 percentage points more sycophantic than human respondents, with GPT-4o among the most sycophantic despite performing well on explicit sycophancy benchmarks, a dissociation suggesting that standard evaluations fail to capture real-world sycophantic behavior. In a pre-registered experiment with 557 participants performing a Bayesian reasoning task, Batista & Griffiths (2026) demonstrate that default GPT-5.1 is statistically indistinguishable from a rule-confirming prompt (confirmed via equivalence test): both suppress discovery rates to 5.9–8.4%, compared with 29.5% under random feedback. Users rarely detect the distortion: Bo et al. (2026) find that 71% of participants fail to identify sycophantic responses, and overreliance rates reach 47.4% with sycophantic AI compared with 3% under low-sycophancy conditions. Moreover, LLMs affirm both sides of moral conflicts in 48% of cases (Cheng et al., 2025), raising concerns about diachronic value drift that extend beyond factual accuracy. Attempts to correct sycophancy face a design dilemma: anti-sycophancy overcorrection replaces affirmation bias with different pathologies such as presumptuous judgment (Jacobowitz, 2026), and even newer models average roughly one sycophantic behavior per conversational turn (Kirgis et al., 2026). The relationship between personalization and sycophancy is role-dependent: advisory roles increase epistemic independence while peer-framed roles decrease it (Kelley & Riedl, 2026). Sun & Wang (2026) identify a subtler mechanism ($N=224$): neutral-toned LLMs that adapt their stance are perceived as *more* authentic than complimentary ones, meaning agreement without overt flattery increases trust precisely because it is perceived as genuine. The individual links documented above—sycophancy increasing trust, trust increasing reliance, reliance degrading skills, and degraded skills reducing error-detection capacity—form a self-reinforcing cycle (Jacobowitz, 2026; Maynard, 2026); while each link is individually supported, the loop as a coupled dynamical system has not been directly tested.

6.3 Collective and Societal Consequences

The individual-level distortions documented above do not remain individual. As AI-assisted work becomes routine across professions, these effects aggregate into changes in collective knowledge production, creative diversity, and the professional pipelines that produce future domain experts.

Epistemic infrastructure under pressure. Kwon (2026) formalize how individual-level distortions scale through a mathematical model of AI-accelerated scientific writing, predicting a “deceptive honeymoon” in which publication volume surges while review quality silently degrades, so that net scientific knowledge output peaks briefly before declining well below baseline. Knowledge communities that traditionally served as epistemic infrastructure are already being displaced: Burtch et al. (2024) show that ChatGPT’s release reduced activity on Stack Overflow, disproportionately among newer contributors, though Reddit communities, where social bonds provide value beyond information retrieval, were buffered against the same displacement, suggesting that purely informational knowledge ecosystems are most vulnerable. At a deeper level, Guingrich et al. (2026) distinguish “belief offloading” from cognitive offloading (Risko & Gilbert, 2016): while the latter delegates computation, the former delegates belief formation itself, concentrating epistemic power in the few providers whose models generate the beliefs users adopt. The pattern extends to scientific peer review, where 6.5–16.9% of sentences show signs of AI modification, correlated with lower reviewer confidence and deadline pressure (Liang et al., 2024), suggesting that evaluative effort is already shifting at the population level.

AI boosts individual quality but narrows collective diversity. When Si et al. (2024) had an LLM ideation agent independently generate research ideas, the AI-generated ideas were rated higher in novelty than expert-written ones on average, yet only approximately 200 unique ideas emerged from 4,000 seeds per topic (~5%), indicating that LLMs channel output toward a narrow region of idea space. Ashkinaze et al. (2024) find that high passive AI exposure increases the diversity of ideas within a population but does not improve individual creativity; AI makes ideas *different from each other*, not *better*. In writing, Jo & Raghavan (2026) demonstrate through a pre-registered RCT ($N=200$) that AI-assisted drafts converge in style and content, though human editing substantially recovers diversity, and originality incentives further increase it; in visual creative output, low prompt originality drives homogenization through feedback loops in which engagement rewards aesthetic conformity rather than novelty (De Rosa Palmini & Cetinic, 2024). Two findings temper the homogenization narrative. First, Cheng & Zhang (2025) identify dual mechanisms (inspiration for simple tasks but fixation for complex ones), suggesting that task complexity mediates the direction of AI’s creative influence; Kumar et al. (2026b) demonstrate in two pre-registered experiments ($N=315$ and $N=247$) that homogenization is a deployment artifact, not an inherent cost of AI assistance: a single LLM increases idea-level similarity relative to a no-AI control ($p=.033$), but pairing two architecturally distinct LLMs eliminates the effect entirely ($p=.950$ vs. control). Second, Wang et al. (2026e) document a temporal trajectory in which student journalists initially defer to AI-generated content but gradually reassert creative independence, suggesting that initial convergence need not be permanent.

The missing junior loop. AI adoption shifts the nature of work from production to evaluation. Simkute et al. (2024) characterize this as a “production-to-evaluation shift” in which the human role narrows to quality judgment over AI-generated output; Bauer et al. (2025) provide a theoretical grounding, categorizing AI effects along an inversion–redefinition continuum in which unscaffolded use undermines deep conceptual learning while redesigned tasks can foster it. Randazzo et al. (2025) identify three interaction modes among 244 management consultants: Cyborg (60%, tight human-AI integration on each subtask), Centaur (14%, strategic delegation of distinct subtasks), and Self-Automator (27%, near-complete delegation with minimal review). That more than a quarter of skilled professionals largely delegated judgment, despite working under review, signals that structural adoption without parametric internalization operates at the organizational level, not only in laboratory settings. Catalini et al. (2026) formalize the resulting “Missing Junior Loop”: if AI substitutes for junior work, the apprenticeship pipeline that produces future experts erodes, creating a delayed competence deficit invisible in short-term productivity statistics. Chen et al. (2025d) document a concrete instance in programming education ($N=24$): beginners with ChatGPT access achieve five-fold higher task scores yet show markedly weaker conceptual understanding, with complete-solution generation yielding the largest performance–understanding gap. A systematic review of AI in medical education corroborates the pattern: across four studies ($N=408$), AI improves efficiency and basic knowledge but produces mixed or inferior outcomes for higher-order clinical reasoning (Turney et al., 2026). A labor-market review reports that one study found AI-adopting U.S. firms reduced entry-level hiring (ages 22–25) by 13% (Rio-Chanona et al., 2025), consistent with the pipeline contraction that the missing junior loop predicts.

These collective consequences—eroding epistemic infrastructure, narrowing creative diversity, and contracting the pipeline that produces future evaluators—are not simply individual effects scaled up. They are qualitatively distinct: the missing junior loop, for instance, degrades evaluative capacity not in current users but in the next generation of domain experts who may never acquire the competence that apprenticeship would have provided.

6.4 Design Principles for Parametric Internalization

Whether AI use drives structural adoption alone or also enables parametric internalization depends on the selective pressures that shape human adaptation; that is, on whether human-side change is self-driven and unmanaged or steered by externally designed interventions. Brynjolfsson et al. (2025) provide the clearest evidence that genuine internalization is possible: among customer service agents, approximately 15% productivity gains (36% for the lowest-skilled workers) persisted through an AI system outage, demonstrating durable skill transfer rather than mere tool dependence. One especially promising approach augments human intermediaries rather than end-users directly. In a pre-registered RCT with 900 tutors and 1,787 students, Wang et al. (2024c) show that an AI copilot assisting *tutors*, not students, produces a 4 percentage point gain

in student mastery (9 percentage points for students of lower-rated tutors), at a cost of \$20 per tutor per year; the mechanism is that treatment tutors shift toward higher-quality pedagogical strategies rather than simply relaying AI-generated answers. More broadly, when systems give humans flexible control over delegation, humans voluntarily retain more tasks: [Feng et al. \(2026a\)](#) find that a co-planning system interleaving human and agent execution steps yields higher steerability than a chat baseline, and researchers in a field deployment assign more steps to themselves when given delegation control. Three design principles emerge from the broader literature, each supported by controlled experiments.

Scaffolding over supplanting. When AI is designed to scaffold reasoning rather than supply answers (a principle rooted in Vygotsky’s zone of proximal development ([Vygotsky, 1978](#))), learning gains are substantial. [Kestin et al. \(2025\)](#) find that a custom pedagogically-designed AI tutor outperforms in-class active learning by 0.73–1.3 standard deviations in a crossover RCT ($N=194$). [Su et al. \(2026b\)](#) show that a Socratic chatbot elicits 73% more interactions and 40% greater cognitive diversity than a general-purpose chatbot, even when final task performance does not differ, indicating that the scaffolded design improves the *process* of learning, not only its outcome. Convergent results appear in Socratic tutoring for research-question formulation ([Degen & Asanov, 2025](#)); [Lee et al. \(2024\)](#) find in a 10-week RCT ($N=61$) that a guidance-based ChatGPT wrapper requiring students to attempt answers before receiving hints significantly improves cognitive engagement, critical thinking, and knowledge retention on delayed tests ($p=.006$) relative to unrestricted ChatGPT. The contrast with direct-answer AI is stark: the same studies that document learning harm under unguided access (e.g., the -17% exam effect in [Bastani et al., 2025](#)) show that scaffolded variants eliminate or sharply reduce the deficit; [Dasari et al. \(2024\)](#) demonstrate the magnitude of this gap in a three-group quasi-experiment ($N=29$): ChatGPT-only students score $M=9.8$ versus $M=42.6$ for teacher-plus-ChatGPT, confirming that unsupported AI access yields dramatically worse learning outcomes. [Oakley et al. \(2025\)](#) provide a neuroscience-grounded account: internalized knowledge (engrams, schemata, neural manifolds) is structurally required for higher-order thinking and error detection via prediction errors, making the decoupling a consequence of bypassing knowledge internalization, not merely reduced practice time. Yet a systematic review of 21 studies in music education finds that empowerment and dependence coexist even within scaffolded designs, with no evidence that current generative AI tools support forethought or specific self-regulatory sub-processes ([Peng et al., 2026a](#)), a qualification suggesting that scaffolding is necessary but not sufficient for genuine parametric internalization; structured workflows that ground AI co-writing in the learner’s own ideas can partially address this gap ([Ding et al., 2025](#)). [Güner & Er \(2025\)](#) find convergent evidence in a quasi-experiment ($N=158$): five distinct AI interaction profiles emerge, ranging from passive code generation to independent coding, and instructional interventions (prompting training, guided sample prompts) significantly shift students toward collaborative profiles, with more independent profiles associated with higher post-test scores.

Deferred access. Requiring users to attempt a task before receiving AI assistance produces better outcomes than immediate access. [Singh et al. \(2026\)](#) demonstrate in a three-condition RCT ($N=97$) that a write-then-revise condition, where students draft hints before seeing AI-generated alternatives, yields the highest-quality outputs on error-identification tasks, whereas immediate AI access encourages substitution. This principle aligns with the broader finding that the “vicious cycle” of submit-incorrect-code $\rightarrow$ request-AI-fix $\rightarrow$ resubmit dominates when AI is available from the start ([Rahe & Maalej, 2025](#)), and with radiology trainees’ preference for on-demand over always-visible AI support ([Savardi et al., 2025](#)) and physical procedural tasks ([Bellos et al., 2025](#)).

Epistemic friction. Deliberately deploying weaker or less fluent models can promote critical evaluation. [Islam et al. \(2026\)](#) find that a reflection-first protocol (requiring students to draft their own analysis before consulting a local 7B-parameter model) produces significant downward shifts in AI credibility perceptions, prompting independent verification behaviors. The underlying mechanism, reducing processing fluency to counteract metacognitive blindness, suggests that *less* capable AI may sometimes be pedagogically superior; a related proposal has AI deliberately feign confusion to force learners to diagnose errors rather than consume answers ([Tomisu et al., 2025](#)). [Ghosh et al. \(2026\)](#) find convergent evidence with a different friction mechanism ($N=214$): an “Assumptions” cognitive forcing function that prompts users to identify implicit premises in AI-generated plans achieved the lowest overreliance rate (63.0% vs. 80.7% in the comparison

condition, $p=.008$), without increasing cognitive load. A complementary approach modulates support based on user trust state: [Srinivasan & Thomason \(2026\)](#) demonstrate across five sequential experiments ($N>500$) that trust-adaptive explanations (supportive framing when trust is low, counter-explanations when trust is high) substantially reduce inappropriate reliance.

These three principles share a common logic: each preserves the cognitive engagement that direct-answer AI suppresses, creating conditions under which structural adoption can coexist with parametric internalization rather than displacing it ([Singh et al., 2025](#)). Yet they face a fundamental tension: meaningful oversight of AI outputs may require performing the very task that was delegated. [Mezzadri \(2025\)](#) formalize this *verification paradox*: quality control of AI-generated research artifacts (literature reviews, data summaries) demands the domain competence that AI use was meant to supplement, implying that efficiency gains and epistemic integrity pull in opposite directions. [Tang et al. \(2024\)](#) document a concrete instance: in an eye-tracking study ($N=28$), developers aware of LLM code provenance fix substantially more bugs but exhibit elevated fixation durations and self-reported effort, confirming that meaningful verification of AI output imposes a measurable cognitive cost. The tension is not merely theoretical. Institutional frameworks lag behind practice: the EU AI Act names automation bias as the only cognitive bias warranting regulatory attention but provides no operational definition of effective oversight ([Laux, 2023](#); [Laux & Ruschemeier, 2025](#)), while judges in Colombia, Peru, Mexico, and India already use ChatGPT in judicial reasoning without established safeguards ([Socol de la Osa & Remolina, 2024](#)). RAG-based legal AI tools hallucinate at rates of 17–33% despite vendor reliability claims ([Magesh et al., 2024](#)), yet the users most likely to adopt these tools, time-pressured professionals, are precisely those least able to verify outputs. Training partially resolves this tension: [Chen & Bao \(2026\)](#) show in an RCT ($N=164$) that a brief training intervention (a 9.5-minute video with quiz) shifts legal AI use from unproductive (untrained users showed no improvement over non-users) to beneficial (+0.27 grade points), suggesting that the binding constraint is often not technology design but user preparation; [Clerc et al. \(2026\)](#) find a parallel result in a different population and domain (middle-school science, $N=116$): a two-hour AI literacy workshop improves LLM-assisted task performance, yet self-reported AI competence does not predict actual performance ($r=0.01$).

The parallel to the agent side is notable: just as the vast majority of agent evolution operates under autonomous selective pressure (§5), the majority of human cognitive adaptation to AI proceeds without institutional design or structured intervention. The evidence indicates that the performance–capability paradox is a selective-pressure problem: it arises when human-side adaptation is self-driven and unmanaged, and it is attenuated when externally designed interventions preserve cognitive engagement.

These design principles are not permanent solutions. The human–AI relationship is non-stationary: optimal AI assistance changes as human experience accumulates ([Jain et al., 2025](#)), AI system behavior shifts over deployment ([Kirgis et al., 2026](#)), and users renegotiate their reliance, sometimes toward independence ([Robertson, 2025](#); [Wang et al., 2026e](#)), sometimes toward deeper integration, whether through genuine learning or automation bias ([Goldsmith-Pinkham et al., 2026](#)). Any fixed design eventually misaligns with both partners’ trajectories; Section 7 examines what this non-stationarity implies for the coupled system.

Key Takeaways: Human Adaptation

- **The performance–capability paradox is the central empirical pattern:** AI reliably improves measurable output (structural adoption, human-side Δ) while frequently degrading the independent capability needed to produce that output (parametric internalization, human-side ∇ ; see §2.3 for agent-side definitions). This decoupling is documented across education, professional work, decision-making, and creative domains.
- **Two factors determine the trajectory:** task structure (well-defined $\rightarrow$ compression; open-ended $\rightarrow$ amplification) and AI design (scaffolded $\rightarrow$ beneficial; direct-answer $\rightarrow$ harmful). Metacognitive blindness may render these dynamics self-reinforcing.
- **The paradox is a design failure, not an inevitability:** Scaffolded AI, deferred access, and epistemic friction each suggest that parametric internalization is achievable; the evidence is strongest when AI augments human intermediaries rather than replacing the human learning process. However, the human–AI relationship is non-stationary, and any fixed design eventually falls out of alignment as both partners change.

Section 7 places these human-side findings alongside the agent-side analysis from Sections 3–5 and examines what emerges from the comparison.

7 Discussion

Sections 3–5 found that agent self-evaluation is unreliable without an external verifier: autonomous selective pressure succeeds where deterministic verifiers exist but degrades where the evaluator is coupled to the system being evolved. Section 6 found that human self-assessment is unreliable without external scaffolding: unscaffolded AI use produces the performance–capability paradox, and metacognitive blindness prevents users from detecting the deficit. The two literatures developed these findings independently—through different mechanisms (evaluator drift on the agent side, metacognitive blindness on the human side) and in different domains—but they point in the same direction: self-assessment fails when the assessor lacks independence from the thing being assessed. This section examines what follows when the two sets of findings are placed side by side.

7.1 Mutual Dependency

These parallel findings create a dependency between the two sides of the coupled system.

Agent evolution beyond the verifier boundary requires human evaluative input. Where Tier 1 verification is unavailable (open-ended dialogue, value-laden decisions, cultural sensitivity) agent evolution either does not occur or relies on self-referential signals that degrade with iteration. Beyond these formal boundaries, a remaining evaluator whose fidelity is not coupled to the system being evolved is a human who can exercise value-sensitive judgment, that is, human-involved selective pressure ($\mathcal{H} \neq 0$) at sufficient bandwidth.

Human evaluative capacity requires deliberate scaffolding. Yet the human who should provide that evaluative input is not a static resource. Section 6 documented that unscaffolded AI use degrades the independent capability required for evaluation: the verification paradox (Mezzadri, 2025) implies that quality control of AI outputs demands the domain competence that AI use erodes, and metacognitive blindness ensures these effects proceed without the user’s awareness. Maintaining human evaluative capacity therefore requires the externally designed scaffolding that Section 6 identified as effective but rare.

The dependency is mutual. Agent evolution beyond the verifier boundary depends on human evaluative capacity (established above); human evaluative capacity depends on scaffolded design (established above). It follows that effective scaffolding requires understanding how humans adapt to AI, and that understanding in turn informs agent design—closing a loop that no paper in the surveyed literature addresses as an integrated whole. Kumar et al. (2025) find that only 10% of 77 co-learning studies validate mutual adaptation in real-world settings; Shen et al. (2025) document across 411 papers that the field overwhelmingly studies how to align AI with human preferences while neglecting how AI reshapes human cognition.

7.2 Reinterpreting Axis III

In the framework of Section 2.4, $\mathcal{H} \neq 0$ treats human input as entering the evolutionary loop from outside the system—implicitly, as an exogenous signal. The dependency analysis reveals that $\mathcal{H}$ is better understood as *endogenous*: the human who provides evaluative feedback at iteration t has been shaped by the system’s outputs at iterations 1 through $t-1$, and Section 6 showed that this shaping is not negligible.

When the analysis follows Axis III to its boundary (the verifier boundary in Section 5), it encounters a question it cannot answer from agent-side evidence alone: is the human signal reliable? Answering that question requires the reverse analysis in Section 6, and the answer (that default interaction erodes the evaluative capacity on which human selective pressure depends) transforms the framework’s understanding of its own third axis.

Among the $\mathcal{H} \neq 0$ systems in the survey, four track changes in user facts, preferences, or domain knowledge (Chhikara et al., 2025; He et al., 2025b; Nie et al., 2026b; Rasmussen et al., 2025), but none tracks whether the user’s evaluative capacity has been preserved, the level at which the vulnerability operates. This reinterpretation does not invalidate the $\mathcal{H} = 0$ / $\mathcal{H} \neq 0$ classification (the distinction remains operationally meaningful) but it changes what the classification implies: $\mathcal{H} \neq 0$ does not guarantee effective evaluative input, because the quality of the human signal is itself a function of the evolutionary trajectory it is meant to guide.

7.3 Deliberate Design Outperforms the Default

The findings above might suggest a pessimistic conclusion, that the coupled system is structurally destined for degradation. The evidence does not support this interpretation. On both sides, deliberate design outperforms the unmanaged default.

On the agent side, hybrid $\Delta + \nabla$ systems tend to outperform single-pathway alternatives (Section 4), and the few $\mathcal{H} \neq 0$ systems that provide human selective pressure at sufficient bandwidth address domains where deterministic verifiers are unavailable (Section 5). On the human side, the scaffolded designs documented in Section 6 produce genuine parametric internalization: pedagogically designed tutoring yields large learning advantages (Kestin et al., 2025), intermediary-augmentation models that assist tutors rather than replacing student cognition produce measurable mastery gains (Wang et al., 2024c), and skill transfer from AI-assisted work persists through system outages (Brynjolfsson et al., 2025).

The contrast is consistent across both sides: where design provides an independent check (an external verifier for agents, a scaffolded interaction for humans) the outcome is more likely to be durable capability gain. Where design ignores the constraint, the default trajectory produces signal collapse on the agent side and the performance–capability paradox on the human side.

The mutual dependency established above means the coupled system is not two independent optimization problems that happen to share a boundary. Agent evolution shapes the environment in which human evaluative capacity develops; human evaluative capacity shapes the selective pressure on which agent evolution beyond the verifier boundary depends. The two trajectories are coupled, and steering them requires treating agent evolution and human adaptation as components of a single co-evolving system.

7.4 From Self-Improving Agents to Co-Evolving Systems

The mutual dependency identifies what is missing from current practice. The five forms of human-involved input defined in Section 2.4 provide a vocabulary for what the transition requires.

The bandwidth gap. The $\mathcal{H} \neq 0$ papers concentrate in the two lowest-bandwidth forms: implicit signal and evaluative feedback (Section 5). Demonstration appears in one paper, interactive collaboration in two, and direct prescription as a live evolutionary input in none. The higher-bandwidth forms that would allow sustained human steering of agent evolution in value-laden domains remain almost entirely unexplored. This matters because the domains where agent self-evaluation fails (open-ended dialogue, cultural sensitivity, fairness) are precisely those where richer human input would be most relevant—yet the reinterpretation above showed that the quality of that input cannot be taken for granted.

Bridging two timescales. Agent evolution accumulates over timescales from minutes to months; human adaptation unfolds over weeks to years (Robertson, 2025; Goldsmith-Pinkham et al., 2026). No mechanism in either literature tracks both trajectories jointly. Agent reward models encode a snapshot of human preferences that grows stale as the human adapts (the frozen-preference problem identified above); meanwhile, optimal AI assistance shifts as human skill develops (Jain et al., 2025), so a scaffolding design that works at deployment may not work six months later. Current agent benchmarks do not test whether an agent’s evolutionary trajectory degrades the human evaluator; current human-AI studies do not test whether the AI changed between sessions. Neither captures the coupled trajectory that the mutual dependency implies.

What joint evaluation would require. Evaluating a co-evolving system cannot be reduced to evaluating each side independently. On the agent side, it requires tracking not only the evolved agent’s performance but whether the evaluation signal itself remains calibrated across iterations—a form of meta-evaluation that no paper in the surveyed corpus proposes. On the human side, it requires longitudinal measurement: nearly all studies span a single session to one semester, yet the non-stationarity evidence implies that the most consequential effects emerge only over extended deployment. The missing junior loop (Catalini et al., 2026) operates on a still longer timescale—the contraction of apprenticeship pipelines that produce future domain experts is invisible in any single-semester study.

Current systems optimize one side while treating the other as fixed; the evidence reviewed here shows that neither side is fixed, and that their trajectories are coupled. The transition from self-improving agents to co-evolving human–AI systems requires designing systems that monitor and maintain both partners’ capacities jointly, over the timescales at which each actually changes.

Key Takeaways: Discussion

- **Parallel findings:** Agent self-evaluation is unreliable without an external verifier (evaluator drift); human self-assessment is unreliable without external scaffolding (metacognitive blindness). Two independent literatures converge on this from opposite directions.
- **Dependency:** Agent evolution beyond the verifier boundary requires human evaluative input; human evaluative capacity requires scaffolded design; neither side can be solved in isolation.
- **Reinterpretation:** $\mathcal{H} \neq 0$ is endogenous: the human who provides selective pressure is reshaped by the evolutionary process those signals are meant to guide. The framework’s analysis changes the framework’s understanding of its own axis.
- **Co-evolution:** Current systems optimize one side while treating the other as fixed. The transition from self-improving agents to co-evolving systems requires closing the human-input bandwidth gap, bridging the timescale mismatch between agent iterations and human adaptation, and building joint evaluation infrastructure that tracks both partners over deployment lifetimes.

8 Conclusion

The three-axis framework revealed that agent evolution and human cognitive adaptation are coupled—each side depends on the other, and the coupling is largely unmanaged.

Section 7.4 identified the transition requirements at the coupling level—bandwidth, timescale, and joint evaluation. The coupling also operates at the level of agent-internal design choices that the earlier axis-by-axis analysis characterized independently. Three connections, each spanning the agent-side and human-side analyses, illustrate what a co-evolutionary perspective would change.

First, consolidation pathway choice is not only an agent-performance trade-off but a human-verifiability trade-off. Structural consolidation produces artifacts that are human-readable and auditable (prompts, code, workflow graphs; Section 4); parametric consolidation produces weight changes that are not. If human evaluative capacity is degrading under default interaction (Section 6), then parametric-dominant evolution compounds an analogous verification challenge (Mezzadri, 2025) by removing a primary channel through which a human evaluator could inspect what changed, while structural-dominant evolution at least preserves inspectability. No current system selects its consolidation pathway based on whether the human partner can verify the result.

Second, different evolutionary substrates may affect different dimensions of human cognition. Section 3 classified agent evolution by substrate (Cortex, Action, Memory); Section 6 classified human adaptation along corresponding dimensions (cognitive, operational, epistemic). Indirect evidence suggests the distinction matters: studies of reasoning-assistive AI document declines in deliberative capacity (Section 6), while studies of information-retrieval AI document shifts in epistemic habits such as belief offloading (Guingrich et al., 2026) and knowledge-community displacement (Burtch et al., 2024). Whether these differential effects can

be traced to the evolutionary substrate of the AI system—and which combinations of agent evolution and human scaffolding are most critical to co-design—remains open.

Third, the hybrid $\Delta+\nabla$ systems that Section 4 identified as the rapidly emerging pathway category deserve investigation as candidates for co-evolving systems, precisely because they maintain a human-readable structural component alongside a parametric component. Whether the structural component can serve not only as an agent-internal knowledge representation but as an interface for human oversight is an open design question.

Limitations. These conclusions are bounded by several limitations. The three-axis framework assumes static substrate boundaries, yet emerging self-modifying architectures blur the distinctions on which the taxonomy depends. The coupled-system analysis assumes a dyadic setting—one human, one agent—while deployed systems increasingly involve multi-agent architectures and multiple human collaborators whose network dynamics this framing does not capture. The classification was conducted by the authors using a structured rubric without formal external inter-rater reliability assessment. Roughly nine in ten of the surveyed agent-evolution papers were published in 2025–2026, so the taxonomy predominantly reflects a two-year window of a rapidly evolving field. The literature search was conducted primarily in English-language venues, and human-side studies draw disproportionately from WEIRD populations; both sets of findings may differ in underrepresented contexts. Finally, the framework’s own logic applies to itself: if self-assessment fails when the assessor lacks independence from the system being assessed, then this survey’s analytical categories, shaped by researchers who use the same AI tools the survey examines, are subject to the same vulnerability we document.

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